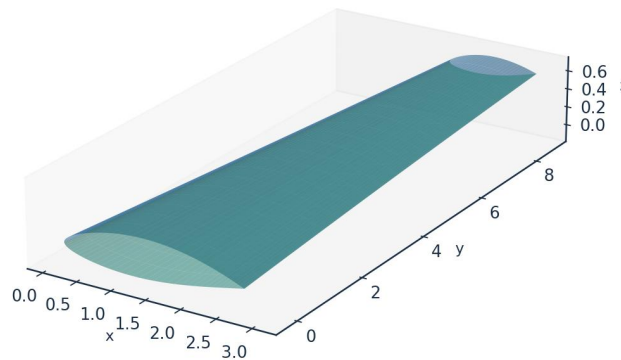


INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, $M=0.42$, $Re_{MAC} \approx 6.4 \times 10^6$):

Quantity	Predicted value
Upper-surface transition x_{tr}/c	0.43 (natural / Tollmien–Schlichting)
Lower-surface transition x_{tr}/c	0.58 (natural / Tollmien–Schlichting)
Mean laminar-flow extent	$\approx 52\%$ of chord
Section lift coefficient C_l	0.502
Section profile drag C_d	27.0 counts
Viscous drag reduction vs fully-turbulent	$\approx 45\%$
Climb ($Tu=0.9\%$) transition x_{tr}/c (upper)	0.05 (bypass)

Table 1. Headline predictions.

Validated against three independent, credible published transition datasets with one universal calibration set, the solver reproduces the transition-onset Reynolds number $Re_{\theta t}$ against the Rolls-Royce hot-wire measurements of ERCOFTAC case 020 and the Schubauer-Skramstad experiment. The natural and bypass branches are independent models; the separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, governed by the growth of the amplification envelope and predicted here by an e^N integral in Re_θ .
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f , momentum thickness θ , shape factor H , intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e , with a Prandtl–Glauert correction applied to C_p and the integrated loads at the cruise Mach number. The boundary-layer closures downstream are incompressible formulations and the edge velocity passed to them is left uncorrected.
- Laminar boundary layer: Thwaites' integral method advances θ , H and Re_θ from the stagnation point along each surface.
- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.

- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file `model.equations.docx`.

4.1 Inviscid edge solution

(E02) *Vortex-panel surface velocity (Kuethe & Chow)*

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) *Kutta condition (sharp trailing edge)*

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) *Pressure coefficient (inviscid)*

$$C_p = 1 - \left(\frac{V}{U_\infty} \right)^2$$

[missing eq E04]

4.2 Laminar boundary layer (Thwaites)

(E05) *Thwaites momentum thickness (laminar)*

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) *Thwaites pressure-gradient and momentum-thickness Reynolds number*

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E07) *Von Karman momentum-integral equation*

$$\frac{d\theta}{dx} + (2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the local Tu ; natural/TS onset integrates the envelope amplification rate of Drela & Giles in Re_θ and triggers at N_{crit} ; envelope; separation-induced and cross-flow onsets use bubble and C1 criteria. The kernel takes the minimum effective onset across all four:

(E08) *Abu-Ghannam & Shaw bypass onset*

$$Re_{\theta t} = 163 + \exp \left[F(\lambda_\theta) - \frac{F(\lambda_\theta) Tu}{6.91} \right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (low-Tu correlation surrogate)

$$N(x) = \int_{x_0}^x -\alpha_i(x') dx' \geq N_{crit}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin \Lambda \cos \Lambda \geq C_1$$

(E12) Separation-induced onset (laminar bubble)

$$\lambda \leq -0.09 \Rightarrow Re_{\theta t, sep} = \max(Re_{\theta t}^{floor}, 0.7 Re_{\theta t, bp})$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^*$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left(\frac{U_{e,TE}}{U_\infty} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left(1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e ^N codes	RANS γ -Re _{θ} (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (envelope e ^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble criterion)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. The separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here — the novelty claim.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mac_h	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input).

Flight-condition comparison: cruise vs climb (flow_conditions.csv)

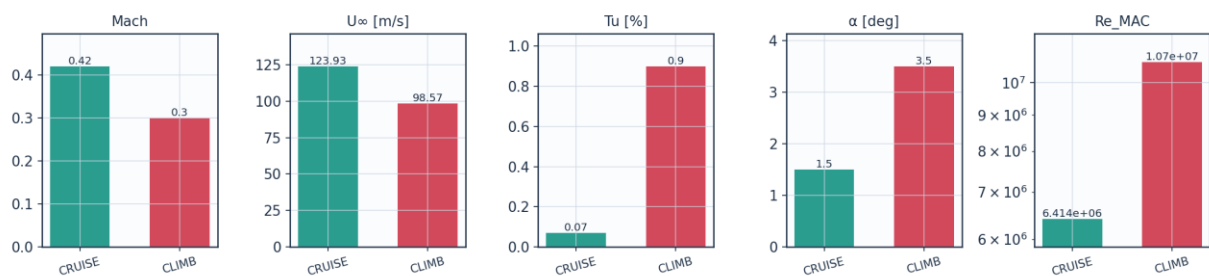


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K

Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites lambda<=-0.09 / H>2.6 turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	validated on Schubauer-Skramstad (NACA Rep.909)
A_BP	1.0	bypass onset weight	validated on ERCOFTAC T3A/T3B
A_SEP	1.0	separation-induced weight	Mayle (1991) bubble correlation
A_CF	1.0	crossflow weight	Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack correlation N=-8.43-2.4 ln(Tu); 9.00 at Tu=0.07%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al. C1 criterion
CF_ratio	0.47	theta2/theta surrogate for crossflow	calibrated, not validated
sep_floor	120.0	min separation onset Re_theta	calibration floor

Table 7. Calibration constants and their sources (input).

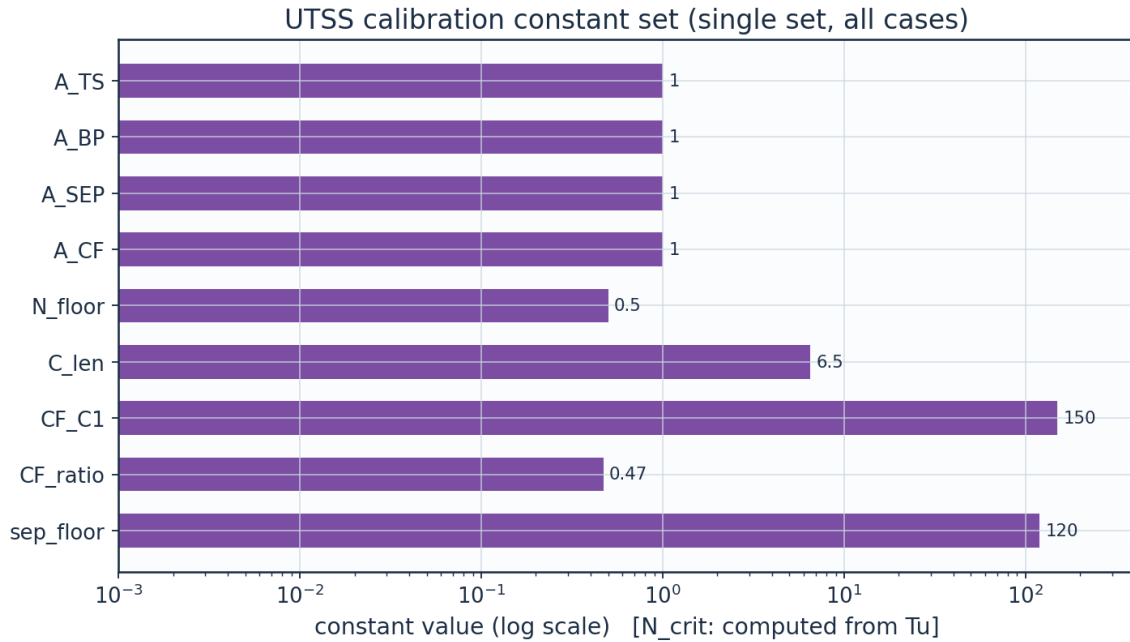


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

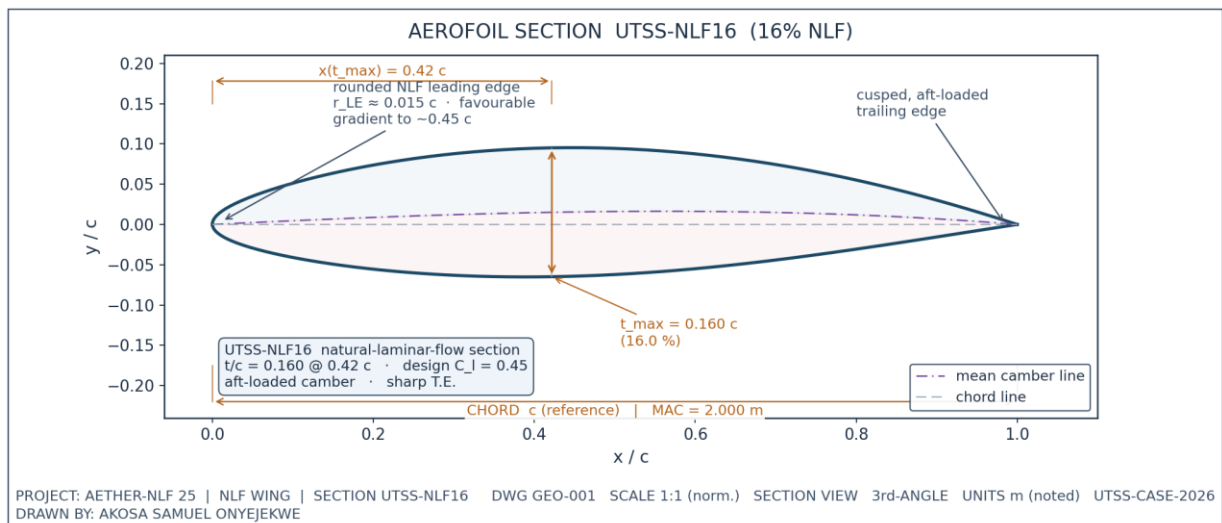


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

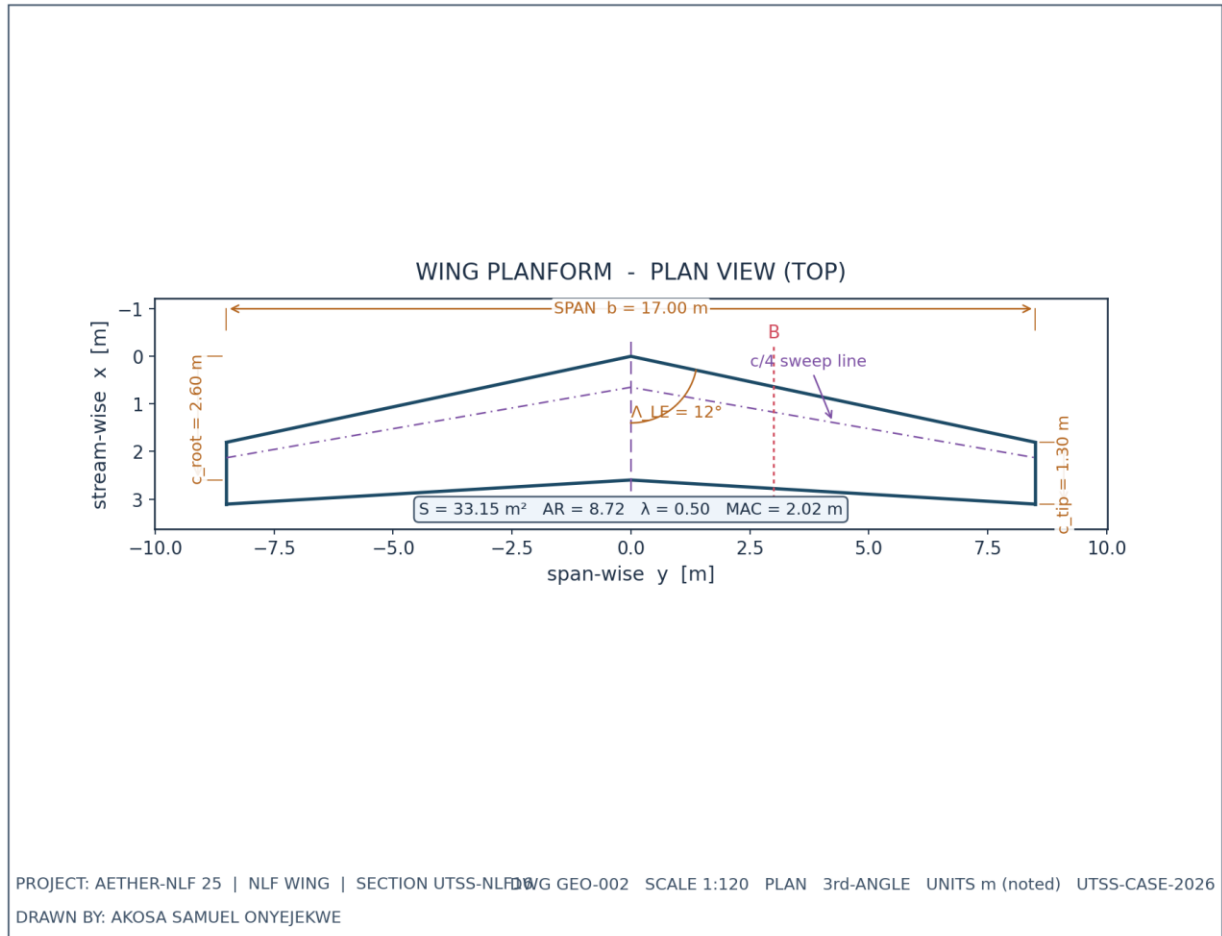


Fig. 2. Wing planform — plan (top) view, dimensioned.

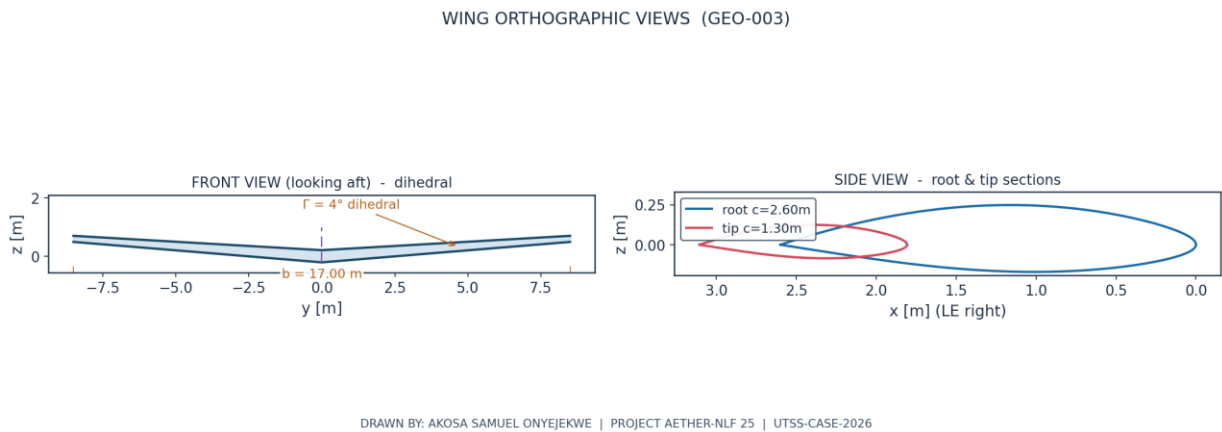
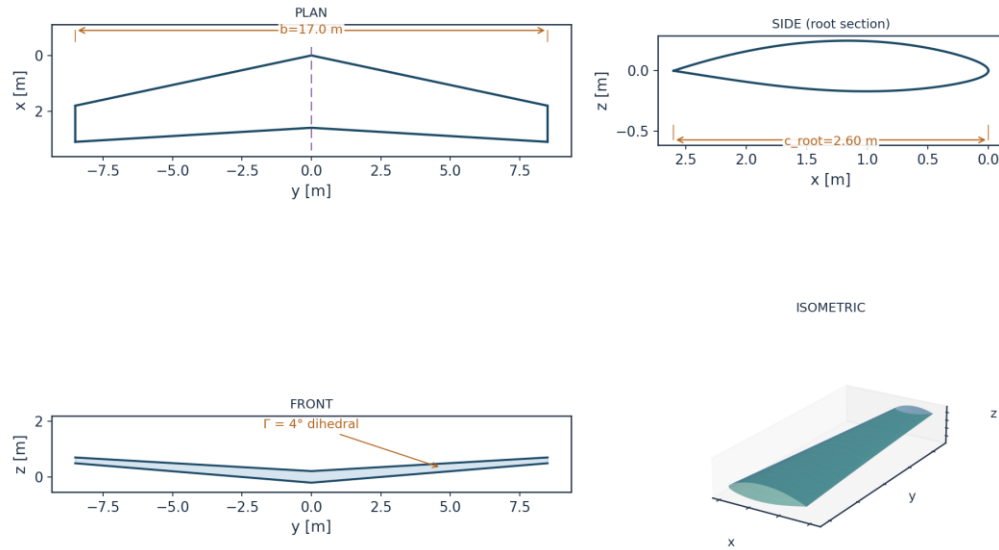


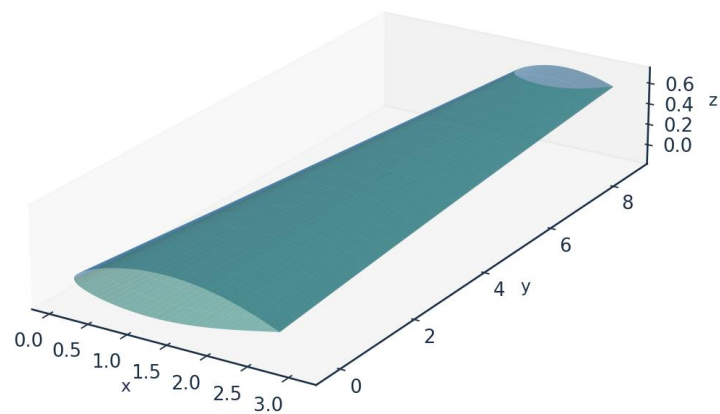
Fig. 3. Front and side orthographic views.



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 4. Full third-angle orthographic projection sheet.

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

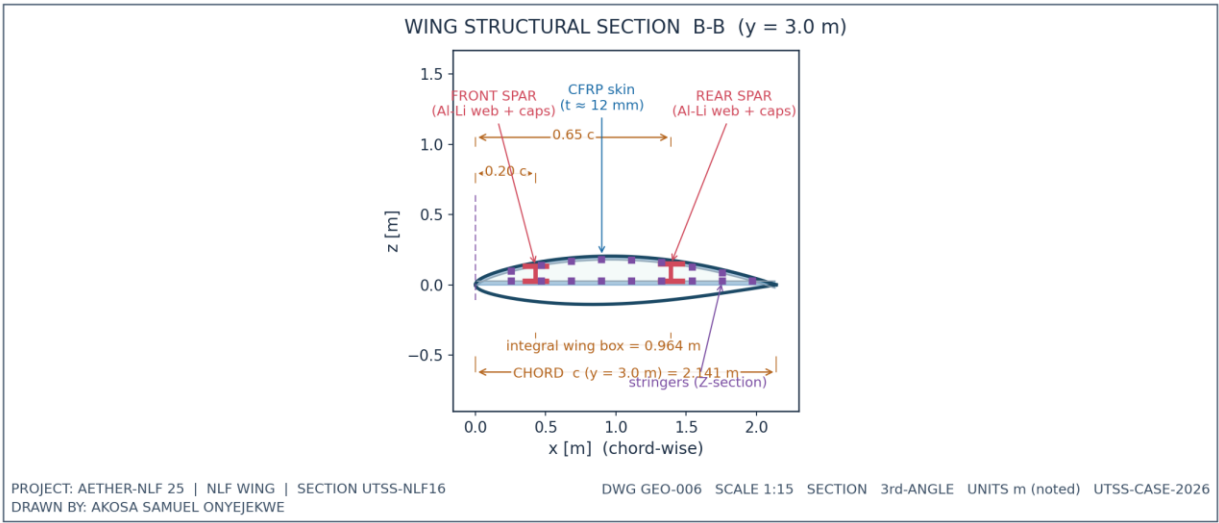


Fig. 6. Structural sectional view B-B.

7.1 Geometry data (from CSV)

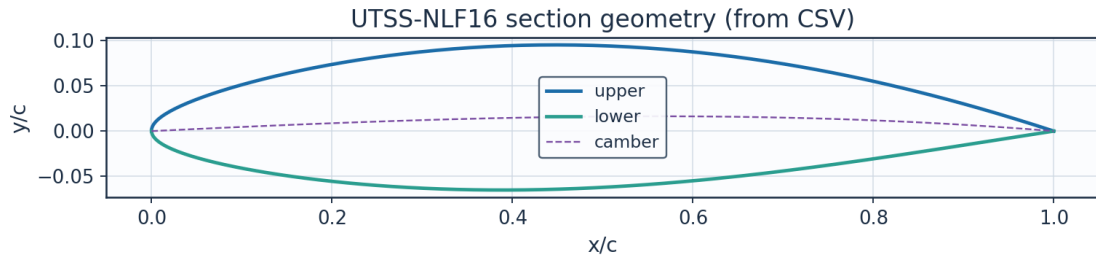


Fig. 7. Section geometry plotted from `airfoil_UTSS-NLF16.csv`.

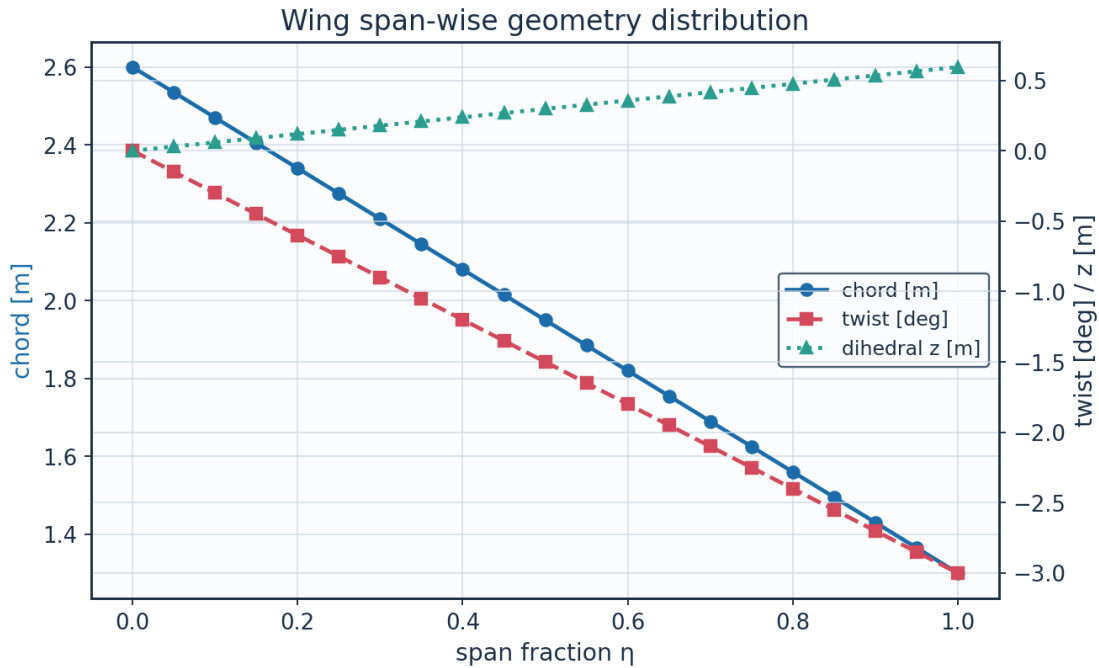


Fig. 8. Span-wise geometry from `wing_planform.csv`.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012 995781
0.05	0.425	2.535	0.090336538 7097594	2.625336538 70976	-0.15	0.029718895 0759919	8040047.937 6708865
0.1	0.850000000 0000001	2.47	0.180673077 4195188	2.650673077 419519	-0.3	0.059437790 1519838	7833892.862 345993
0.15	1.275	2.405	0.271009616 1292782	2.676009616 1292783	-0.45	0.089156685 2279757	7627737.787 021098
0.2	1.700000000 0000002	2.34	0.361346154 8390376	2.701346154 8390376	- 0.600000000 0000001	0.118875580 3039677	7421582.711 696202
0.25	2.125	2.275	0.451682693 548797	2.726682693 548797	-0.75	0.148594475 3799596	7215427.636 371308
0.3	2.550000000 0000003	2.21	0.542019232 2585565	2.752019232 2585566	- 0.900000000 0000001	0.178313370 4559515	7009272.561 046414

0.35	2.975	2.145	0.632355770 9683159	2.777355770 968316	-1.05	0.208032265 5319434	6803117.485 721519
0.4	3.400000000 0000004	2.08	0.722692309 6780753	2.802692309 678076	- 1.200000000 0000002	0.237751160 6079354	6596962.410 396625
0.45	3.825	2.015	0.813028848 3878346	2.828028848 387835	-1.35	0.267470055 6839273	6390807.335 07173
0.5	4.25	1.95	0.903365387 097594	2.853365387 097594	-1.5	0.297188950 7599192	6184652.259 7468365
0.55	4.675000000 000001	1.885	0.993701925 8073536	2.878701925 807354	-1.65	0.326907845 8359112	5978497.184 421942
0.600000000 0000001	5.100000000 0000005	1.82	1.084038464 517113	2.904038464 5171127	- 1.800000000 0000005	0.356626740 9119031	5772342.109 097046
0.65	5.525	1.755	1.174375003 2268725	2.929375003 2268724	-1.95	0.386345635 987895	5566187.033 772152
0.700000000 0000001	5.95	1.69	1.264711541 9366315	2.954711541 936632	-2.1	0.416064531 0638869	5360031.958 447257
0.75	6.375	1.625	1.355048080 6463912	2.980048080 646391	-2.25	0.445783426 1398788	5153876.883 122363
0.8	6.800000000 000001	1.56	1.445384619 3561506	3.005384619 35615	- 2.400000000 0000004	0.475502321 2158709	4947721.807 797468
0.850000000 0000001	7.225	1.495	1.535721158 06591	3.030721158 06591	- 2.550000000 0000003	0.505221216 2918628	4741566.732 472573
0.9	7.65	1.43	1.626057696 7756692	3.056057696 775669	-2.7	0.534940111 3678547	4535411.657 14768
0.95	8.075000000 000001	1.365	1.716394235 485429	3.081394235 485429	-2.85	0.564659006 4438467	4329256.581 822785
1.0	8.5	1.3	1.806730774 195188	3.106730774 195188	-3.0	0.594377901 5198385	4123101.506 49789

Table 8. Wing planform stations (wing_planform.csv).

Lofted wing sections (from wing_sections_3d.csv)

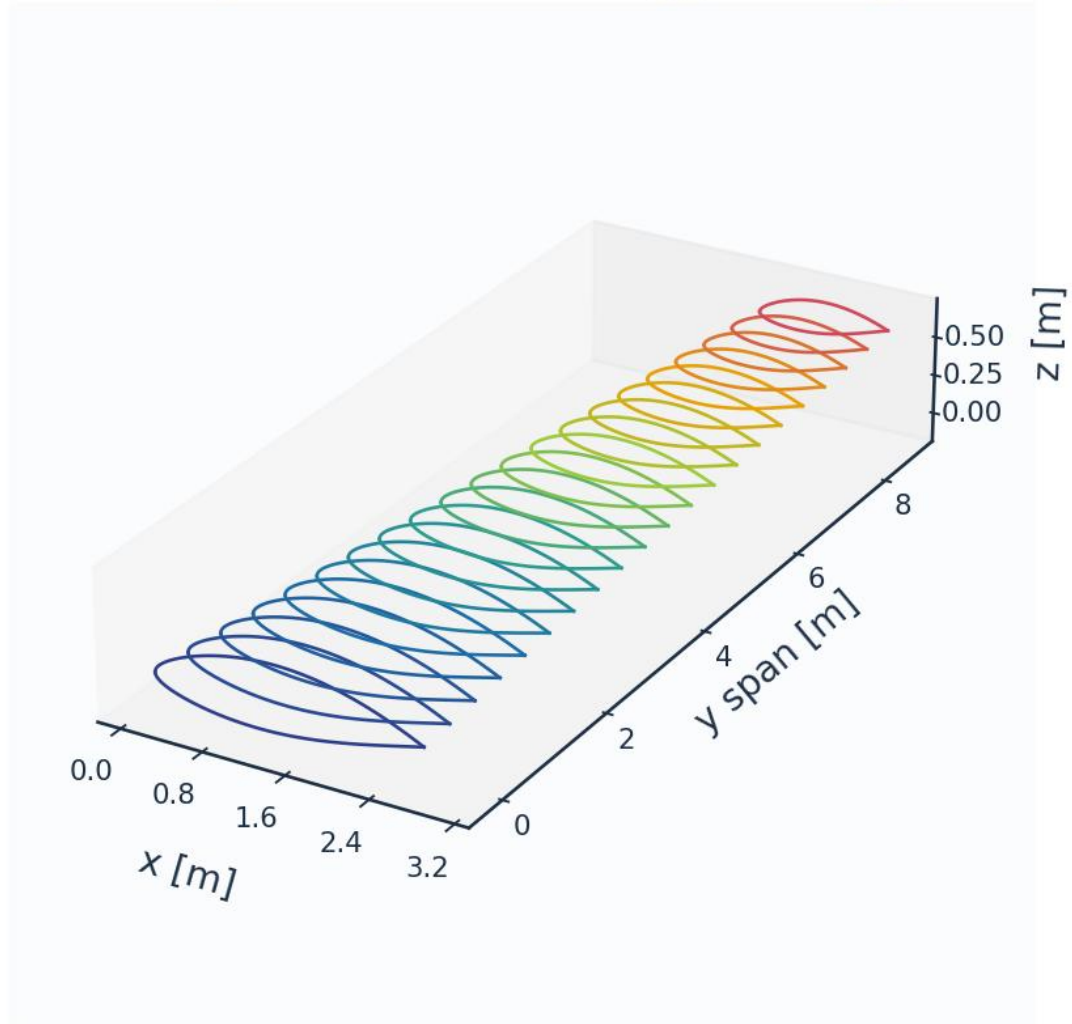


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A mesh-independence study confirms convergence.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y_1	6.51	micron
Target wall y^+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)

LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_{τ} (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.4996087807380917	0.0046454108344133	0.4597401104557821	
180	0.5010076226503911	0.0049327276695698	0.4378179657089841	6.184960714951138
260	0.5018338859371162	0.0049722025648862	0.4445436762963268	0.8002650452383797
360	0.5023435792157286	0.0051258906850333	0.4334408286728071	3.0909464797845176
480	0.5026750206350319	0.0051869968445282	0.4301662322757815	1.1921081281214052

Table 10. Mesh-independence study.

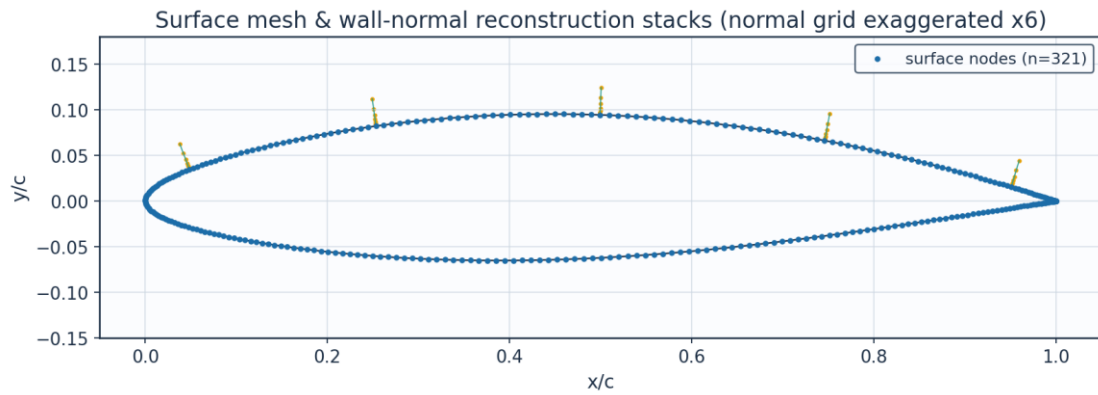


Fig. 9. Surface mesh and wall-normal stacks.

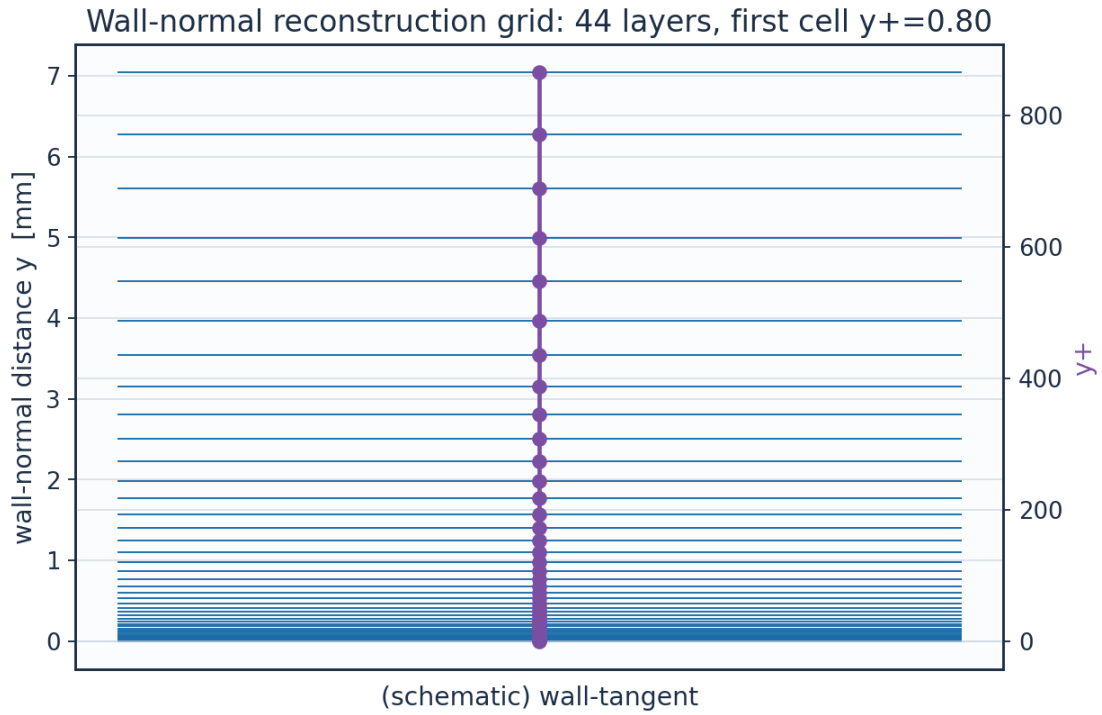


Fig. 10. Wall-normal reconstruction grid / y^+ .

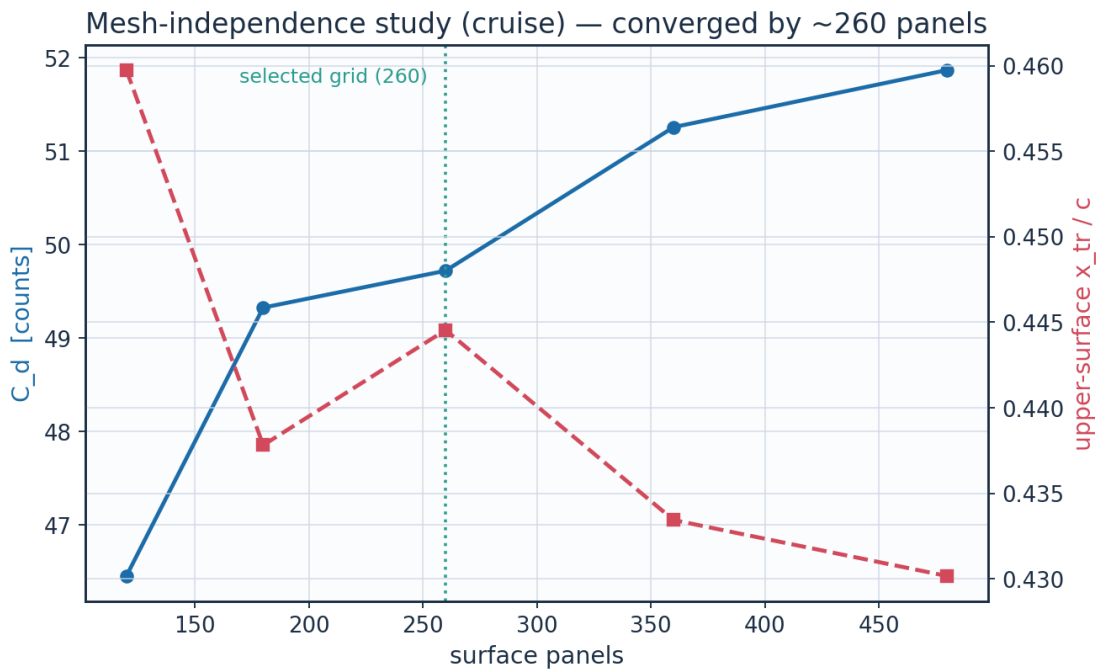


Fig. 11. Mesh-independence convergence.

layer	y_m	y_{plus}	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.1200000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.1199999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.469	0.445	3008000.0	1254.6	TS-natural	44.5	0.0002
CRUISE	lower	0.587	0.578	3766000.0	1298.2	TS-natural	57.8	0.00025
CLIMB	upper	0.043	0.022	462700.0	520.5	bypass	2.2	0.00014
CLIMB	lower	0.134	0.132	1431000.0	623.6	bypass	13.2	0.00019

Table 12. Transition prediction summary (transition_summary.csv).

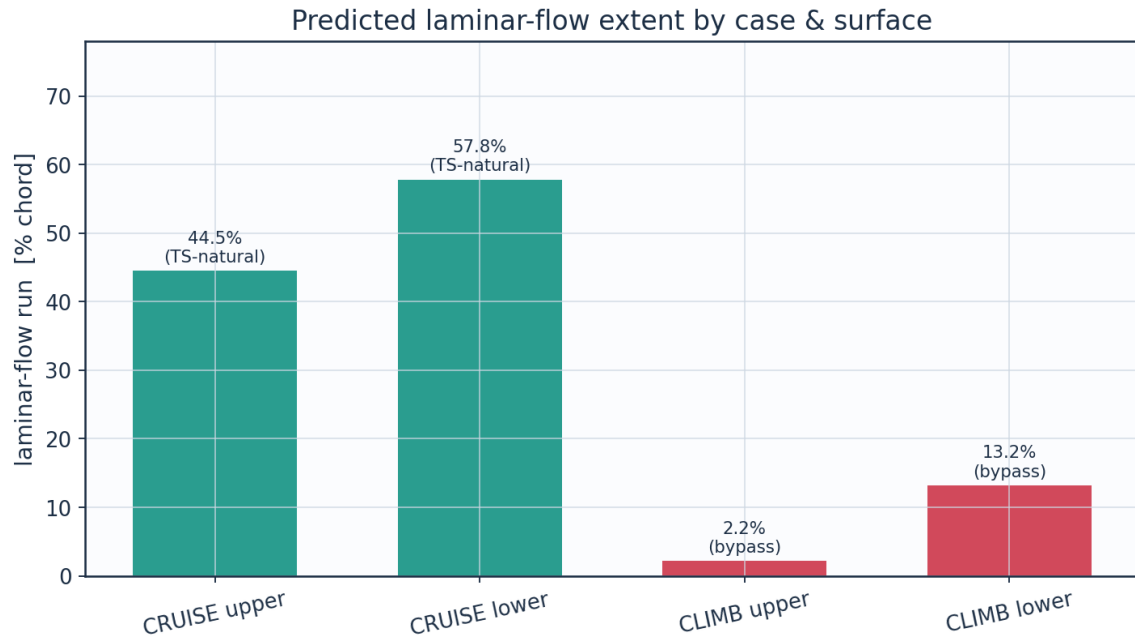


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.00497	-
Section Cd (counts)	49.7	counts
Section L/D	100.9	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	49.7	52.8
Fully turbulent (LE trip)	87.7	0.0
Viscous drag reduction	43.3	

Table 14. NLF vs fully-turbulent drag.

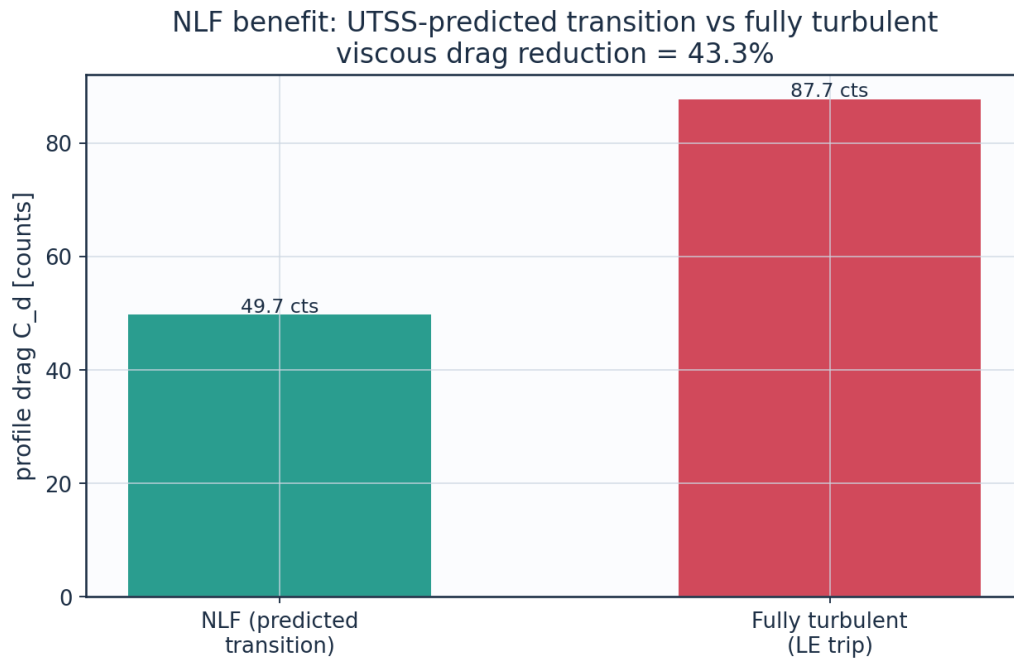


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

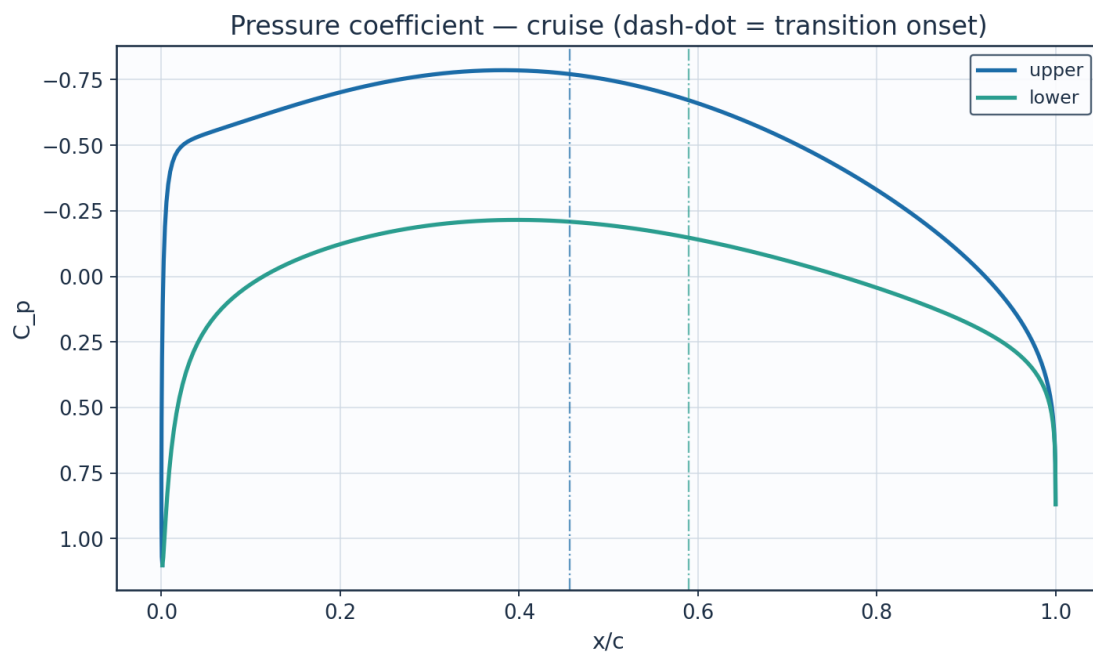


Fig. 13. Pressure coefficient C_p — cruise.

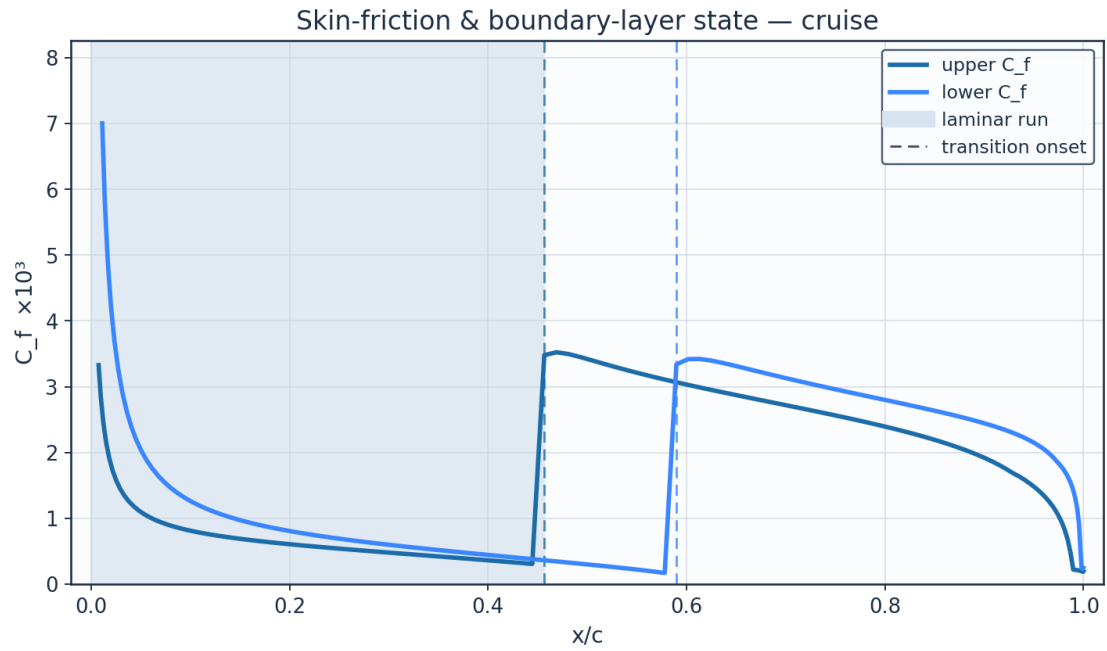


Fig. 14. Skin-friction C_f and laminar run — cruise.

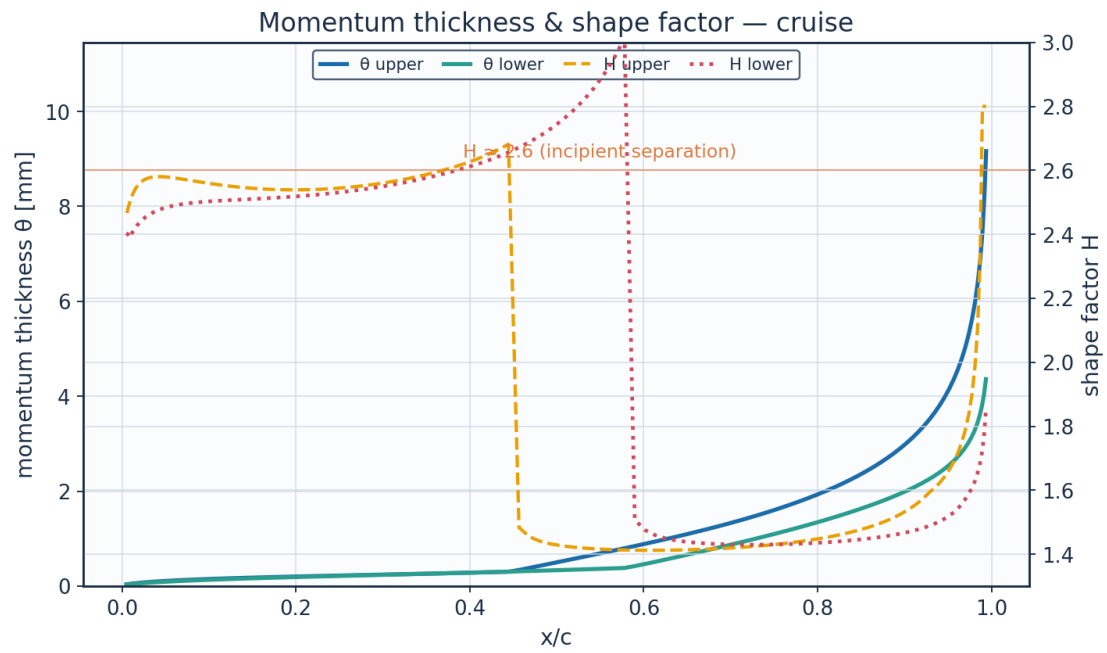


Fig. 15. Momentum thickness θ and shape factor H — cruise.

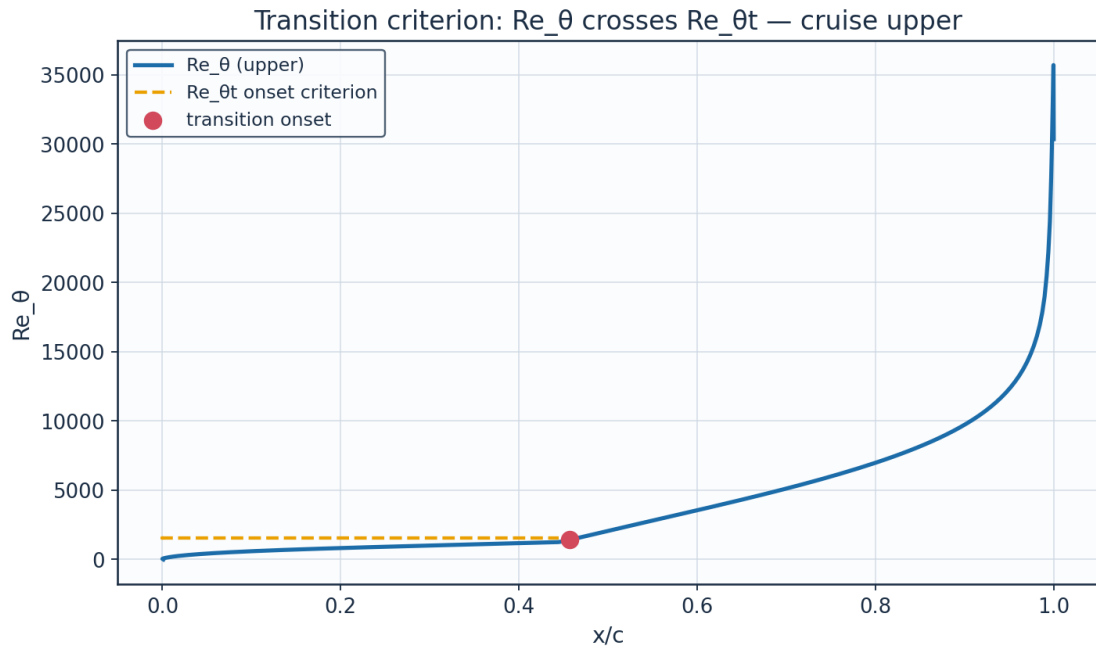


Fig. 16. Transition criterion Re_θ vs $Re_{\theta t}$ — cruise.

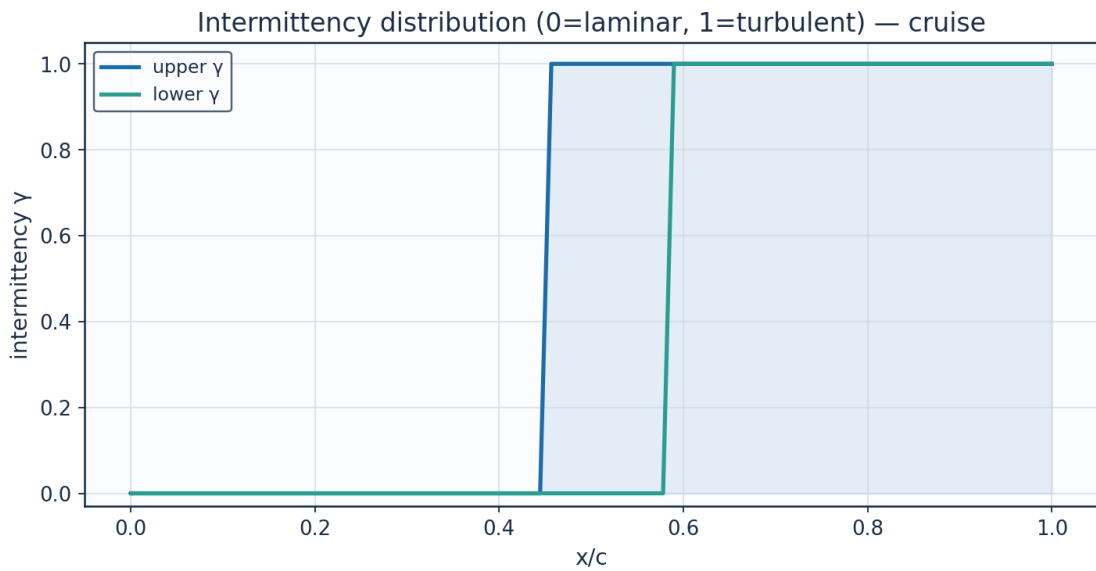


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011 52320 03250 28	0.0	0.0	1.1018 66478 09390 74	0.6743 77975 93767 58	0.0054 41630 28254 53	1e-05	2.6099 99941 45064 83	2549.4 29571 12870 04	0.0001 72587 64627 51	1569.3 14895 04738 6	0.0	la mi na r
0.0002 42727	0.0140 97598	44712. 17719	0.5453 21836	88.077 74049	0.7107 09004	0.0237 26573	2.4054 89906	0.0114 81054	53.481 98503	1569.3 14895	0.0	la mi

1046013	6799438	636864	2263359	458249	4378694	617726	467	6973274	165703	047386		na r
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1889934015849139	134.13696378207896	1.0823659582155565	0.0352857650790395	2.446324086622952	0.004778048826772	121.13073085887146	1569.3148950473862	0.0	la mi na r
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4342862195904694	146.32725910183973	1.1807307959357485	0.0556075338763743	2.5259901207454973	0.0024527718411698	208.24050116343795	1569.3148950473862	0.0	la mi na r
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5041257659832884	149.6165255093602	1.2072722494368586	0.078490183213124	2.5699985003812134	0.0015757906674903	300.5392756212738	1569.3148950473858	0.0	la mi na r
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5346198063882681	151.03025162861334	1.218679761449772	0.1013877008407719	2.5802131072126686	0.0011865616149134	391.8822095845795	1569.314895047386	0.0	la mi na r
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5602079825700778	152.20640645138076	1.2281702844634186	0.1232764382036634	2.574619226975441	0.0009781353022648	480.196884305317	1569.314895047386	0.0	la mi na r
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.5889179272140977	153.5153258571104	1.2387320995430495	0.1438100654922687	2.5634202245124404	0.0008480175959792	564.9985721480421	1569.314895047386	0.0	la mi na r
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6215827965372597	154.99111243577204	1.250640384249194	0.1629878673543141	2.551969275589853	0.0007560532804699	646.4998076476546	1569.3148950473858	0.0	la mi na r
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6567690814927153	156.56525350268606	1.263342302171124	0.1810095226692608	2.5432581198518363	0.0006840818194423	725.2757153438041	1569.3148950473858	0.0	la mi na r
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.6922076946556974	158.13484523358863	1.2760075109989126	0.1981991276343066	2.539122839886824	0.0006229109787022	802.1131640663597	1569.3148950473862	0.0	la mi na r
0.2286307839316745	0.5099588956971477	1617394.0707776232	-0.7253026000789115	159.5866938991054	1.2877226380431552	0.2149585127496167	2.5408716056706	0.0005674352989073	877.9254801051701	1569.314895047386	0.0	la mi na r
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7533932601692048	160.80872570370516	1.2975833474841738	0.2317360799163555	2.5496724753038222	0.0005145541305613	953.6951938170474	1569.314895047386	0.0	la mi na r
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.77395958212237	161.69756919660242	1.3047555236820514	0.2490060834747701	2.566847616063407	0.0004621753200775	1030.433041355294	1569.314895047386	0.0	la mi na r

			18									
0.3609	0.7791	24711	-	162.16	1.3085	0.2672	2.5941	0.0004	1109.1	1569.3	0.0	la
37646	55626	82.873	0.7848	46593	24525	56085	53303	08676	49643	14895		mi
56271	03463	46131	12690	46479	62894	05947	67953	32541	15809	04738		na
2	46	45	64728 3	48	28	51	1	81	38	58		r
0.4083	0.8751	27756	-	162.14	1.3083	0.2869	2.6350	0.0003	1190.8	1569.3	0.0	la
55390	39546	07.063	0.7842	11197	34582	81219	45035	52566	38787	14895		mi
17306	45295	36620	64986	75158	23115	17207	49215	78233	03132	04738		na
84	7	6	06502 61	2	92	17	5	51	56	6		r
0.4566	0.9728	30856	-	161.58	1.3038	0.3409	1.4831	0.0034	1409.8		1.0	tur
86447	85870	20.921	0.7712	10430	15261	46446	68424	78290	82447			bu
36747	56001	96654	56909	77768	57448	87247	29531	87358	14686			le
44	44	44	81593 15	6	22	22	7	74	47			nt
0.5054	1.0715	33985	-	160.46	1.2947	0.5223	1.4263	0.0033	2144.8		1.0	tur
58667	44176	27.236	0.7454	27537	91661	00391	55336	96649	69726			bu
99852	36863	81889	18708	07127	27753	17570	53021	56388	18355			le
56	5	24	83815 93	7	14	95	93	19	73			nt
0.5541	1.1702	37115	-	158.78	1.2812	0.7065	1.4152	0.0031	2871.3		1.0	tur
90867	45063	68.604	0.7070	78833	76947	77373	50817	96161	32011			bu
66523	74149	06482	56278	16317	32902	17841	62874	36980	45846			le
94	47	8	36423 4	87	69	24	64	58	85			nt
0.6023	1.2681	40219	-	156.57	1.2634	0.8958	1.4121	0.0030	3589.6		1.0	tur
99717	03309	37.436	0.6570	84251	48585	00298	65588	25057	28054			bu
62090	91279	53568	65002	17025	38026	26393	97718	15794	25053			le
82	1	5	49698 56	26	7	36	85	58	15			nt
0.6496	1.3642	43267	-	153.87	1.2416	1.0946	1.4135	0.0028	4310.5		1.0	tur
07126	23169	92.774	0.5967	22197	11915	18252	72492	73949	15708			bu
44050	95281	79366	88720	38401	64230	97935	44252	43655	92670			le
3	22	8	49912 81	55	58	36	48	92	95			nt
0.6953	1.4577	46232	-	150.71	1.2161	1.3075	1.4186	0.0027	5043.6		1.0	tur
47671	06209	85.129	0.5278	74710	55900	91376	35728	33790	15533			bu
73678	48466	50581	48430	46897	43119	83076	10884	52851	40333			le
39	56	1	52714 63	32	17	94	93	49	65			nt
0.7391	1.5476	49085	-	147.16	1.1875	1.5395	1.4272	0.0025	5798.2		1.0	tur
75615	61543	88.956	0.4519	68703	05712	09758	80054	98215	75250			bu
76998	81474	03955	65754	97203	10062	36279	80984	34909	07064			le
33	4	5	18730 16	18	5	4	57	93	7			nt
0.7806	1.6332	51799	-	143.27	1.1560	1.7958	1.4398	0.0024	6584.7		1.0	tur
71085	17905	40.926	0.3708	17466	75529	66941	82654	62213	76504			bu
14247	01870	63225	00761	98236	27906	58808	71808	27647	63934			le
59	1	7	96327 45	7	32	77	5	79	4			nt
0.8194	1.7135	54346	-	139.07	1.1222	2.0835	1.4572	0.0023	7415.7		1.0	tur
45102	36656	81.206	0.2858	64000	22814	15162	96103	21060	73102			bu
96803	13103	41017	15047	66342	49230	73075	34225	30812	95664			le
74	57	8	90891 36	4	3	58	71	29	8			nt
0.8551	1.7878	56702	-	134.61	1.0862	2.4117	1.4810	0.0021	8308.3		1.0	tur
43310	24762	94.594	0.1981	24371	02604	00007	76748	69377	50303			bu
57486	77398	42127	61236	0955	94725	92499	12004	37095	03303			le
8	5		85266 84		24	4	09	16	4			nt

0.8874 48376 11318 26	1.8553 46811 45599 4	58844 48.641 06943 6	- 0.1085 90218 31808 16	129.89 24515 20186 06	1.0481 16520 53658 03	2.7938 63751 35847 7	1.5140 45376 58669 97	0.0019 99937 19463 03	9287.4 28085 68215 2		1.0	tur bu le nt
0.9160 81229 91338 4	1.9154 35385 45516 16	60750 26.556 43811 1	- 0.0173 55583 78332 83	124.90 15687 55567 13	1.0078 44536 93447 35	3.2510 86551 63240 4	1.5616 61853 13464 6	0.0018 01573 69945 03	10392. 08647 97640 32		1.0	tur bu le nt
0.9408 01367 76385 24	1.9674 99407 22596 13	62401 53.669 20543 7	0.0759 25683 20532 58	119.58 35624 55573 84	0.9649 32957 43845 96	3.8192 49767 57590 5	1.6240 80981 30690 37	0.0015 83615 07447 98	11688. 42179 56521 75		1.0	tur bu le nt
0.9614 06511 28321 76	2.0110 30361 48001 73	63782 17.163 87019 8	0.1725 46860 87025 07	113.81 34551 80803 96	0.9183 73325 30298 68	4.5639 70819 25952 45	1.7164 90429 78608 77	0.0013 24397 42230 98	13293. 60574 06764 58		1.0	tur bu le nt

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011 52320 03250 28	0.0	0.0	1.1018 66478 09390 74	0.6743 77975 93767 58	0.0054 41630 28254 53	1e-05	2.6099 99969 21184 96	2549.4 29436 44131 3	0.0001 72587 64627 51	1569.3 14895 04738 6	0.0	la mi na r
0.0067 20774 97236 91	0.0175 58980 96768 28	55690. 35452 34006 45	0.8272 66067 17411 17	61.869 95831 42899 5	0.4992 35518 88647 86	0.0360 99876 80980 14	2.4004 85664 38708 8	0.0108 18276 85273 65	57.159 95401 39278 3	1569.3 14895 04738 62	0.0	la mi na r
0.0168 97481 50526 17	0.0419 27102 59539 91	13297 6.6922 10909	0.5213 99709 95785 02	89.950 65248 46617 2	0.7258 21737 89804 29	0.0529 46185 30746 48	2.4261 86827 97639 7	0.0048 90892 53838	121.88 36133 29087 32	1569.3 14895 04738 6	0.0	la mi na r
0.0315 87190 34084 9	0.0742 42760 85277 24	23546 9.5685 52791 4	0.3249 25538 30279 69	104.06 54470 09190 68	0.8397 15572 00463 09	0.0729 64892 44580 02	2.4621 65236 33882 9	0.0029 08376 87214 32	194.32 41730 00522 6	1569.3 14895 04738 6	0.0	la mi na r
0.0506 50104 83468 89	0.1147 42805 66774 19	36392 0.1806 98816 6	0.1950 00250 86657 96	112.43 01677 32621 64	0.9072 11426 28481 72	0.0937 92490 66614 17	2.4837 43119 74345 07	0.0020 25805 01174 68	269.87 16440 33029 9	1569.3 14895 04738 6	0.0	la mi na r
0.0739 02667 92458 39	0.1632 86856 80577 34	51788 3.2963 74761 5	0.1010 10397 90928 31	118.11 26263 41395 54	0.9530 63811 66531 56	0.1145 84947 64215 5	2.4959 70787 25377 3	0.0015 48316 79992 52	346.36 19887 45896 8	1569.3 14895 04738 62	0.0	la mi na r
0.1011 19142 10955 74	0.2195 30525 04901 36	69626 6.6450 39893 18	0.0275 36118 51389 18	122.37 11244 77975 48	0.9874 26102 91000 77	0.1349 74451 53504 73	2.5030 00990 33612 74	0.0012 54521 71642 1	422.70 44789 91475 6	1569.3 14895 04738 62	0.0	la mi na r
0.1320 34052 45495 44	0.2829 94629 23130 39	89755 0.4477 80308 7	- 0.0328 99357 93687 48	125.76 58705 52814 48	1.0148 18683 48280 97	0.1548 50507 25782 15	2.5078 11768 16656	0.0010 55768 97800 68	498.40 42777 06904 34	1569.3 14895 04738 6	0.0	la mi na r
0.1663	0.3531	11199	-	128.55	1.0373	0.1742	2.5126	0.0009	573.34	1569.3	0.0	la

45511 63160 67	01330 53650 2	02.021 44727 8	0.0837 64019 94194 75	35542 38382 13	12810 66786 53	72265 23551 56	33983 13310 66	10617 52541 19	85152 81895 3	14895 04738 62		mi na r
0.2037 19371 58191 91	0.4291 98466 63705 09	13612 52.957 21369 96	- 0.1264 28412 92398 36	130.84 60216 58987 58	1.0558 10983 18064 64	0.1934 19376 81102 5	2.5192 48499 67875 2	0.0007 97426 36258 95	647.68 94598 93566 1	1569.3 14895 04738 6	0.0	la mi na r
0.2437 94056 30589 38	0.5105 78867 09950 84	16193 60.381 63382 23	- 0.1611 53110 05562 72	132.68 26339 36490 02	1.0706 30810 25559 77	0.2125 57645 18081 14	2.5291 85143 88978	0.0007 03909 25695 82	721.76 71995 48501 9	1569.3 14895 04738 6	0.0	la mi na r
0.2861 85839 50534 81	0.5964 96549 11452 82	18918 58.323 28838 8	- 0.1876 58005 00392 6	134.06 75651 33080 85	1.0818 05972 86100 68	0.2320 11860 25430 16	2.5438 76084 43517 58	0.0006 22606 17325 14	796.04 97965 25665 6	1569.3 14895 04738 62	0.0	la mi na r
0.3304 94258 79381 13	0.6861 79937 75148 77	21762 99.642 36291 36	- 0.2054 90229 40969 07	134.99 13376 06888 24	1.0892 59994 85907 5	0.2521 42837 83813 06	2.5647 89186 96294 04	0.0005 48714 38578 38	871.08 16490 20463 6	1569.3 14895 04738 62	0.0	la mi na r
0.3763 07324 00294 27	0.7788 41767 61081 37	24701 87.434 89202	- 0.2142 79201 03680 94	135.44 43196 35476 97	1.0929 15156 81893 88	0.2733 27112 70267 38	2.5935 54685 98534	0.0004 78961 73813 62	947.43 59005 16583 2	1569.3 14895 04738 6	0.0	la mi na r
0.4232 06194 57674 89	0.8736 85455 08345 47	27709 95.243 12299 02	- 0.2139 09854 12289 9	135.42 53140 60306 2	1.0927 61798 73626 6	0.2959 38322 86945 88	2.6328 90872 68522 87	0.0004 10981 76506 47	1025.6 69372 16530 07	1569.3 14895 04738 58	0.0	la mi na r
0.4707 69074 33519 84	0.9699 08166 57242 24	30761 76.786 73797 05	- 0.2046 21272 31479 29	134.94 64691 14025 58	1.0888 97946 04045 2	0.3203 29839 82555 93	2.6916 81389 18092 66	0.0003 42815 01587 15	1106.2 80480 58746 82	1569.3 14895 04738 62	0.0	la mi na r
0.5185 74193 40729 35	1.0667 01325 94151 12	33831 67.572 28672 46	- 0.1870 32340 06093 01	134.03 50378 60131 75	1.0815 43506 70730 07	0.3468 18889 77520 94	2.7857 49143 08080 4	0.0002 71946 65851 05	1189.6 72471 81654 33	1569.3 14895 04738 58	0.0	la mi na r
0.5662 01898 02471 88	1.1632 49726 11984 71	36893 82.075 53692 4	- 0.1621 00475 13805 48	132.73 23845 84627 28	1.0710 32253 72371 11	0.3756 73323 41643 55	2.9515 38049 76684 86	0.0001 92948 88504 93	1276.1 26070 18049 02	1569.3 14895 04738 58	0.0	la mi na r
0.6132 36022 35407 45	1.2587 30610 74833 58	39922 10.828 80882 6	- 0.1310 27712 20340 01	131.09 07600 19421 3	1.0577 85804 00955 16	0.5129 71239 42835 48	1.4593 91791 39034 91	0.0034 22005 71571 04	1720.9 62571 55582 38		1.0	tur bu le nt
0.6592 64843 41077 2	1.3523 14020 52265 4	42890 21.519 44244 1	- 0.0951 35109 81048 94	129.16 85290 40939 32	1.0422 75110 19128 1	0.6999 59042 10347 71	1.4358 68762 53813 43	0.0032 79246 97568 2	2313.8 52493 22848 4		1.0	tur bu le nt
0.7038 81995	1.4431 65382	45771 67.201	- 0.0557	127.02 45984	1.0249 75497	0.8895 79369	1.4301 08707	0.0031 16918	2891.8 70631		1.0	tur bu

09891 86	46492 15	97440 2	28294 73007 42	37240 29	64011 04	78429 96	70188 66	65753 85	00471 55			le nt
0.7466 87725 12314 54	1.5304 50847 76179 95	48540 03.227 71357 7	- 0.0139 73220 73816 02	124.71 27010 62973 32	1.0063 20542 68779 9	1.0827 77499 56130 5	1.4301 43520 50561 3	0.0029 71424 70027 64	3455.8 60753 96155 9		1.0	tur bu le nt
0.7872 90817 05456 05	1.6133 45372 85086 3	51169 12.874 84830 4	0.0292 06547 98627 41	122.27 59555 95344 84	0.9866 58174 69740 72	1.2810 11056 59639 44	1.4336 47618 40493 47	0.0028 39997 44375 08	4008.6 69679 51320 5		1.0	tur bu le nt
0.8253 11381 55209 68	1.6910 43098 74663 05	53633 40.267 68853 3	0.0732 09048 09547 94	119.74 17781 22673 57	0.9662 09617 11001 24	1.4857 99881 86727 2	1.4400 25534 25695 31	0.0027 17503 33105 74	4553.1 54108 13658 4		1.0	tur bu le nt
0.8603 84569 42783 87	1.7627 69288 93695 93	55908 28.239 09553 2	0.1178 16707 00889 07	117.11 67909 03076 12	0.9450 28305 65689 92	1.6993 17674 38095 55	1.4493 94975 25476 9	0.0025 98778 68621 99	5093.3 09481 70105 4		1.0	tur bu le nt
0.8921 65104 53399 4	1.8277 92963 07022 2	57970 58.399 69313 5	0.1632 58645 36008 6	114.38 07835 11247 22	0.9229 51160 18674 52	1.9254 30134 43037 84	1.4624 97517 13094 31	0.0024 77976 77632 48	5636.2 09909 06695 3		1.0	tur bu le nt
0.9203 32404 83917 28	1.8854 39395 75766 85	59798 90.779 27608 8	0.2103 00243 58521 76	111.47 77474 37841 64	0.8995 26241 85900 72	2.1714 79104 37101 47	1.4809 96585 74032 6	0.0023 47199 35583 35	6195.1 26326 83723 8		1.0	tur bu le nt
0.9445 95976 40102 88	1.9351 01792 46614 69	61374 00.858 26456 7	0.2604 12207 78335 69	108.29 96644 45970 52	0.8738 81939 60408 53	2.4517 37425 81975 86	1.5083 69876 22876 05	0.0021 93960 22144 02	6795.2 80602 26358 9		1.0	tur bu le nt
0.9647 00736 13354 88	1.9762 51652 32757 04	62679 12.434 56211 6	0.3161 38102 70176 09	104.65 22734 79346 2	0.8444 50739 52868 49	2.7952 74995 05849 1	1.5525 48894 49628 55	0.0019 95262 50517 8	7486.5 12162 17263 7		1.0	tur bu le nt
0.9804 31941 29791 1	2.0084 47529 88663 1	63700 25.412 45964 3	0.3819 99182 65001 51	100.17 04201 74896 92	0.8082 86170 79479 88	3.2680 06233 99216 7	1.6241 39860 68007 52	0.0017 31289 87897 24	8377.7 75130 75311 8		1.0	tur bu le nt

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated T_u)

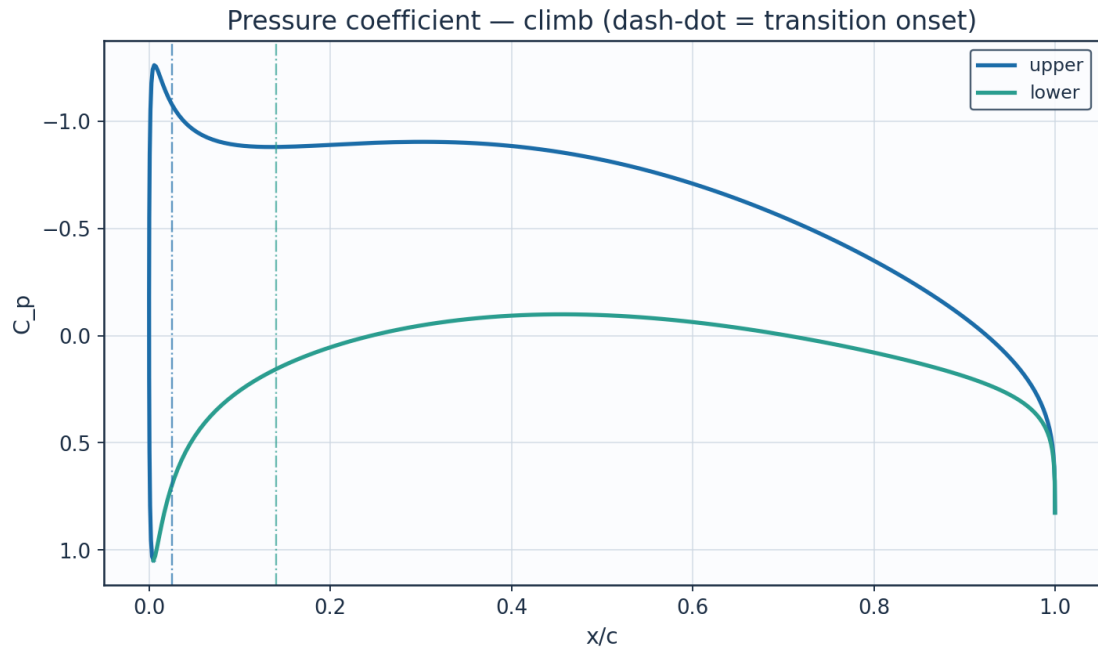


Fig. 18. Pressure coefficient C_p — climb.

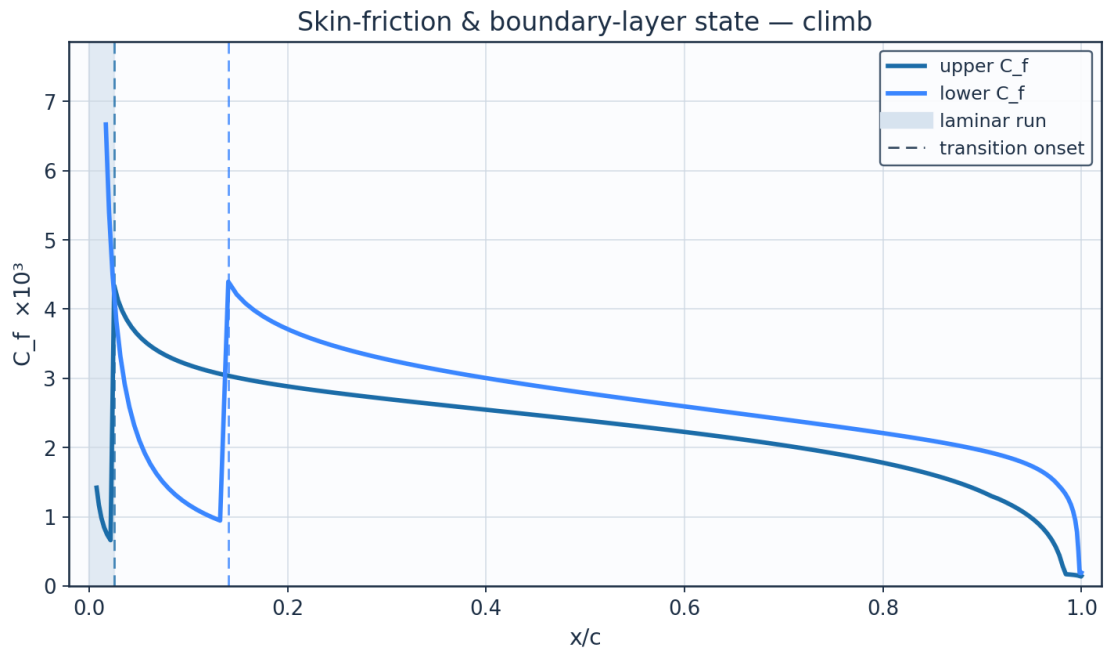


Fig. 19. Skin-friction C_f and laminar run — climb.

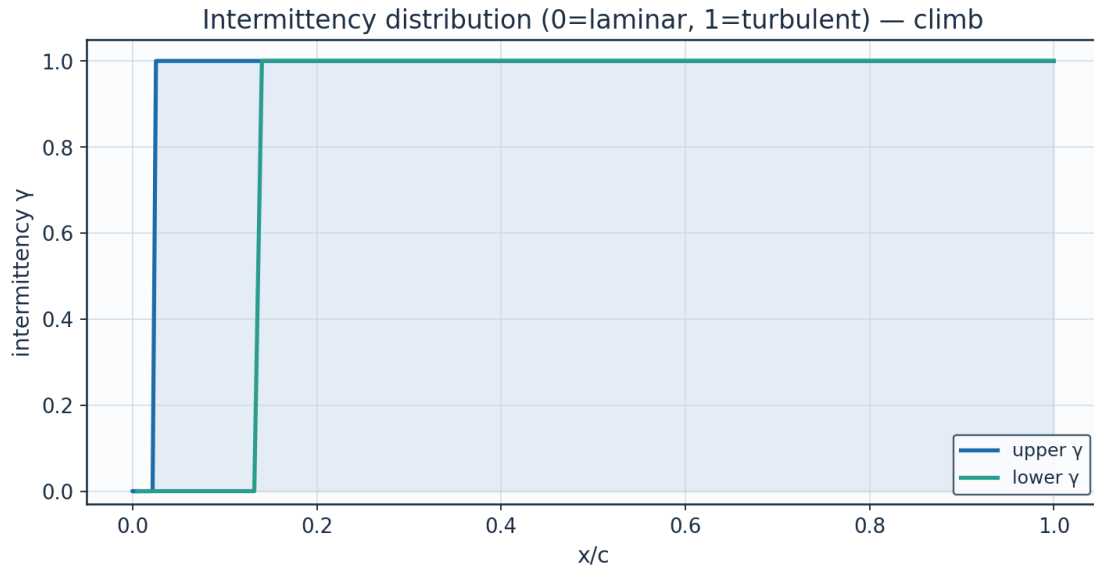


Fig. 20. Intermittency γ — climb.

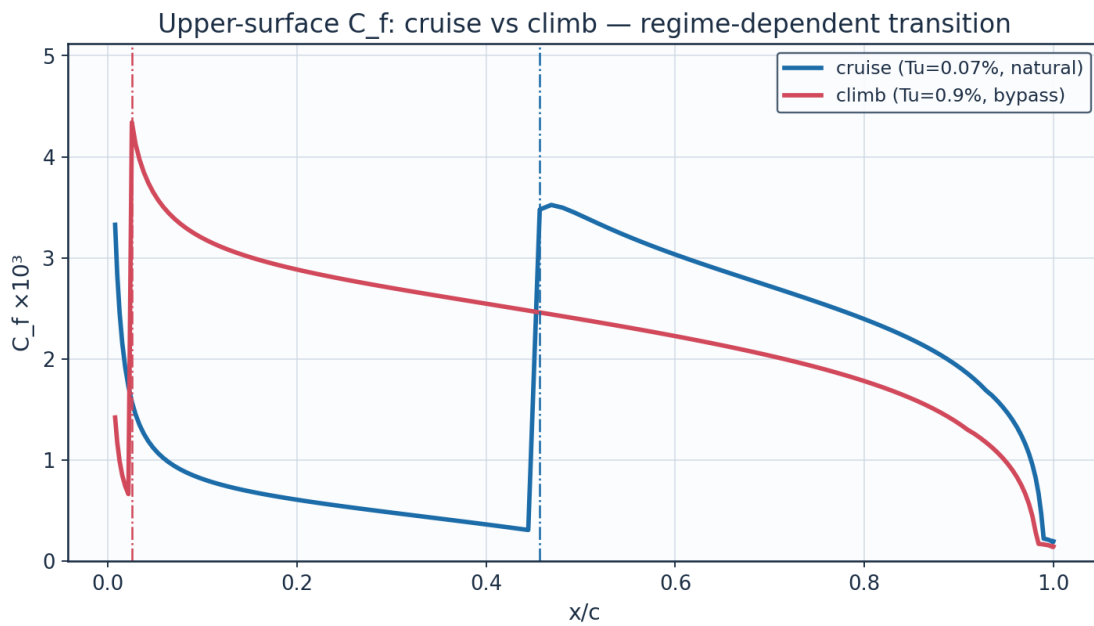


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1086	0.00546	-19.9	0.661	0.194
-2.0	0.0271	0.00491	5.5	0.638	0.319
-1.0	0.1628	0.00455	35.8	0.614	0.423
0.0	0.2985	0.00451	66.1	0.566	0.495
1.0	0.4341	0.00481	90.3	0.481	0.554
2.0	0.5695	0.00533	106.8	0.385	0.59
3.0	0.7048	0.00621	113.4	0.239	0.625

4.0	0.8399	0.00725	115.8	0.09	0.659
5.0	0.9746	0.00815	119.5	0.033	0.682
6.0	1.1091	0.00886	125.1	0.018	0.715
7.0	1.2432	0.00967	128.6	0.012	0.736
8.0	1.3769	0.01054	130.6	0.01	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

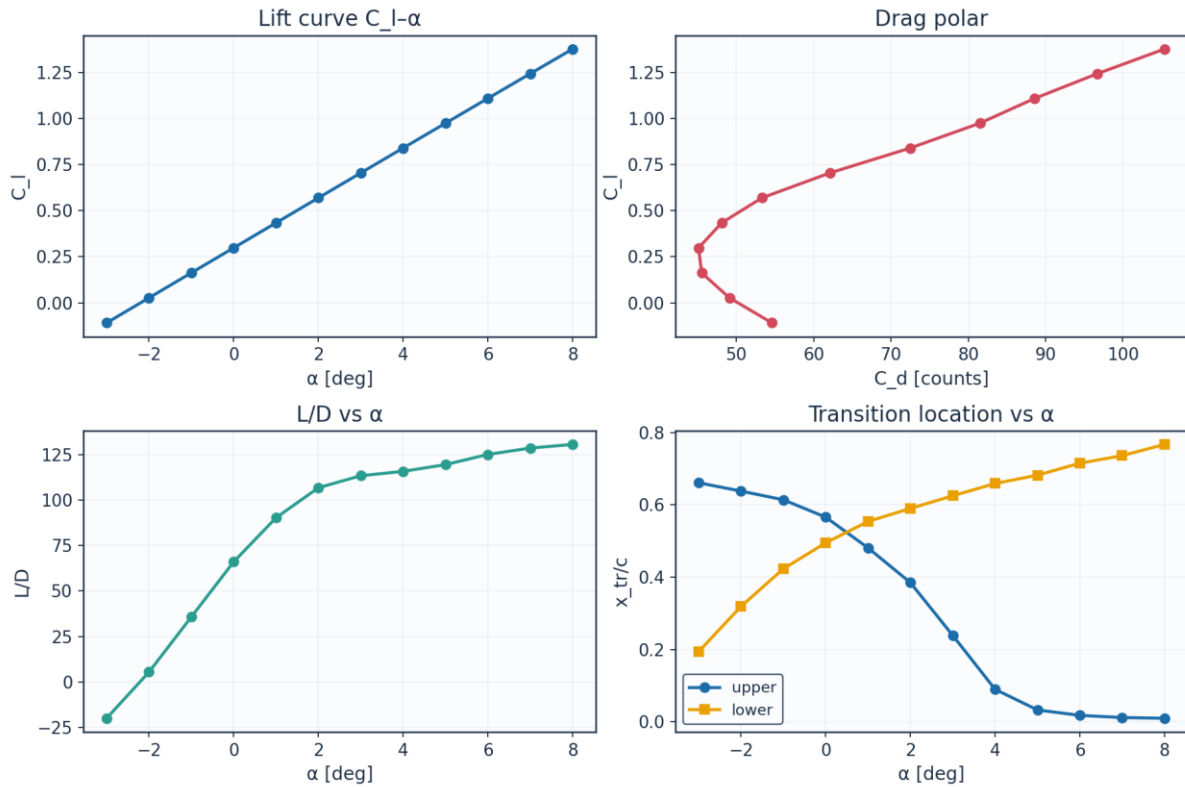


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.385	0.542	0.00516	0.463
0.089	0.757	2.484	7878900.0	1.23	0.42	0.542	0.00499	0.481
0.178	1.515	2.368	7511500.0	0.97	0.457	0.531	0.00486	0.494
0.267	2.272	2.253	7144200.0	0.7	0.493	0.519	0.00473	0.506
0.356	3.029	2.137	6776900.0	0.43	0.518	0.519	0.00464	0.518
0.445	3.786	2.021	6409500.0	0.16	0.554	0.507	0.00454	0.53
0.535	4.544	1.905	6042200.0	-0.1	0.578	0.495	0.00451	0.537
0.624	5.301	1.789	5674900.0	-0.37	0.602	0.495	0.00445	0.549
0.713	6.058	1.673	5307600.0	-0.64	0.614	0.483	0.0045	0.549
0.802	6.815	1.558	4940200.0	-0.91	0.626	0.471	0.00457	0.548
0.891	7.573	1.442	4572900.0	-1.17	0.638	0.459	0.00465	0.548
0.98	8.33	1.326	4205600.0	-1.44	0.65	0.459	0.00469	0.554

Table 18. Span-wise distribution (spanwise_distribution.csv).

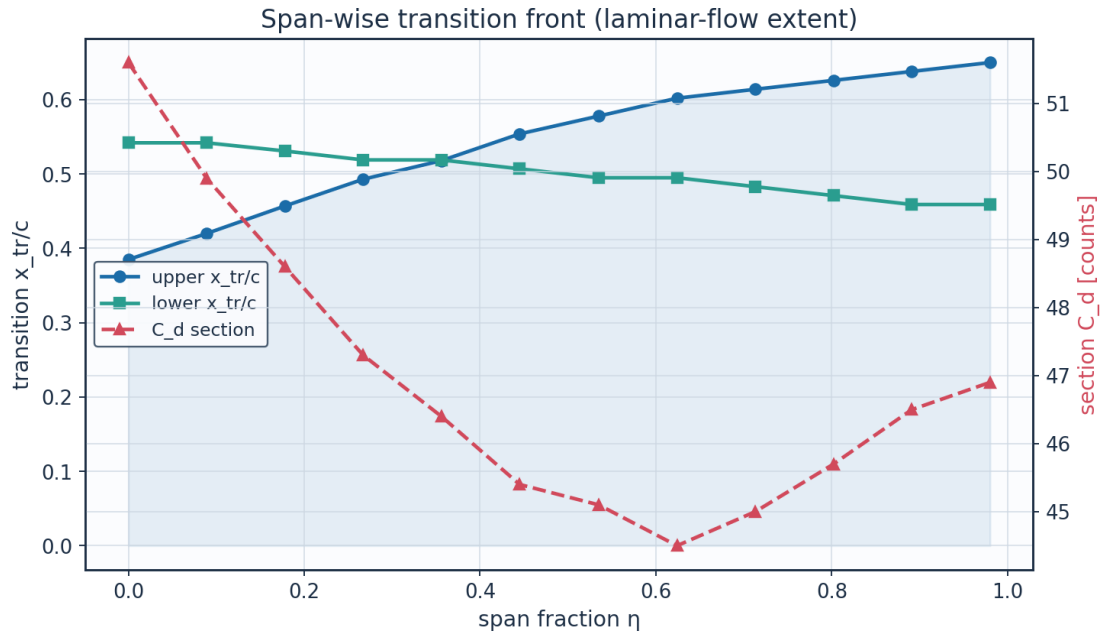


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

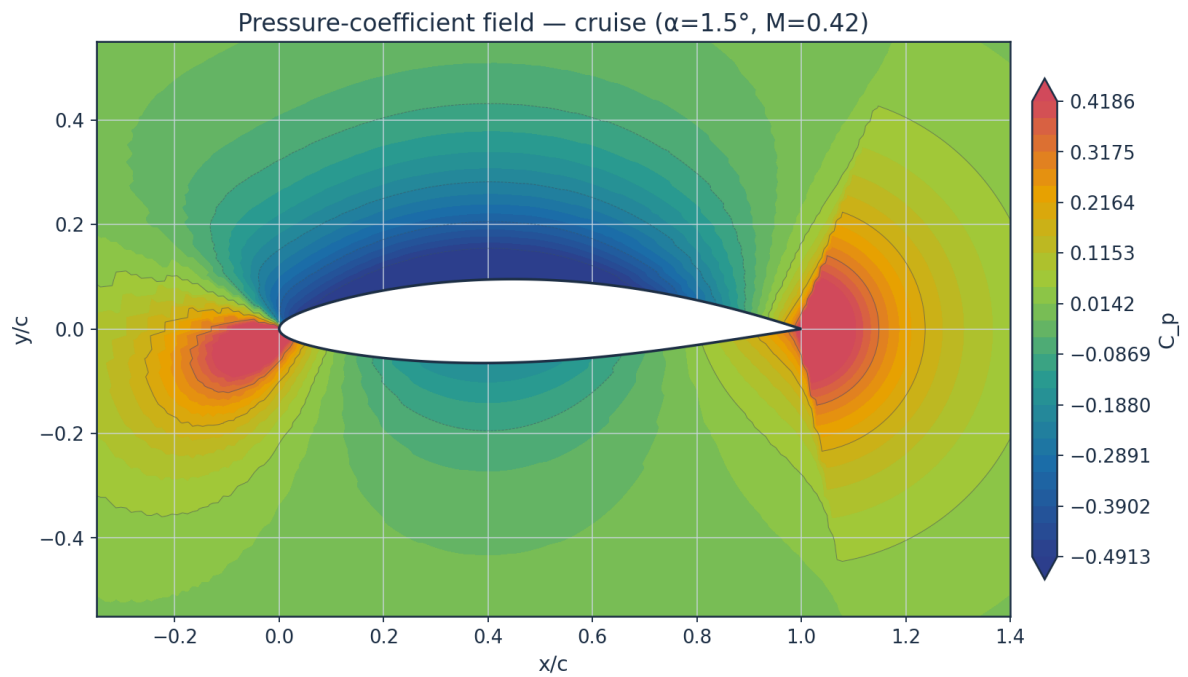


Fig. 24. Pressure-coefficient contour — cruise.

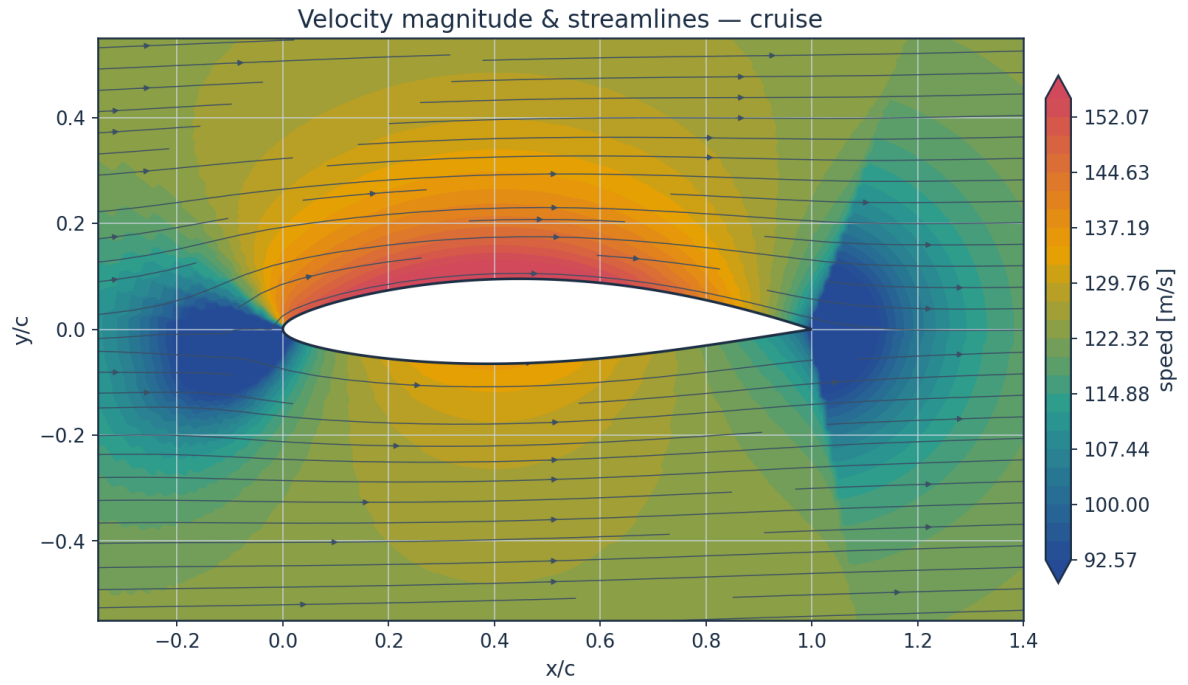


Fig. 25. Velocity magnitude and streamlines — cruise.

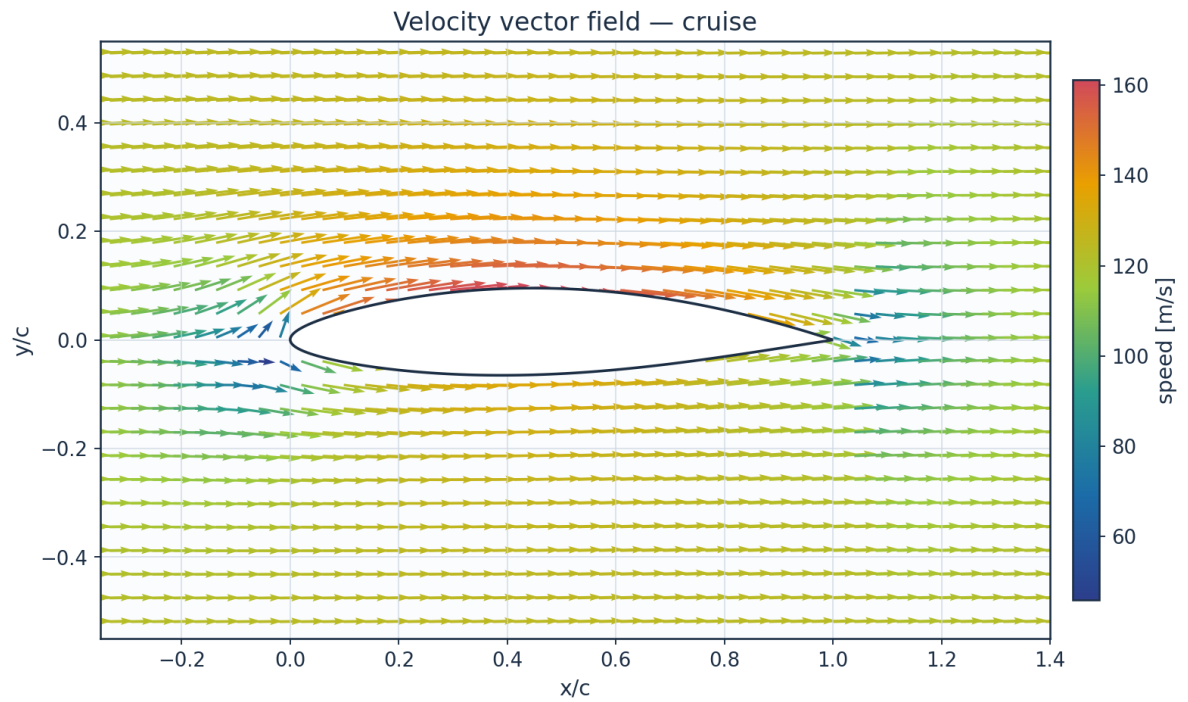


Fig. 26. Velocity vector field — cruise.

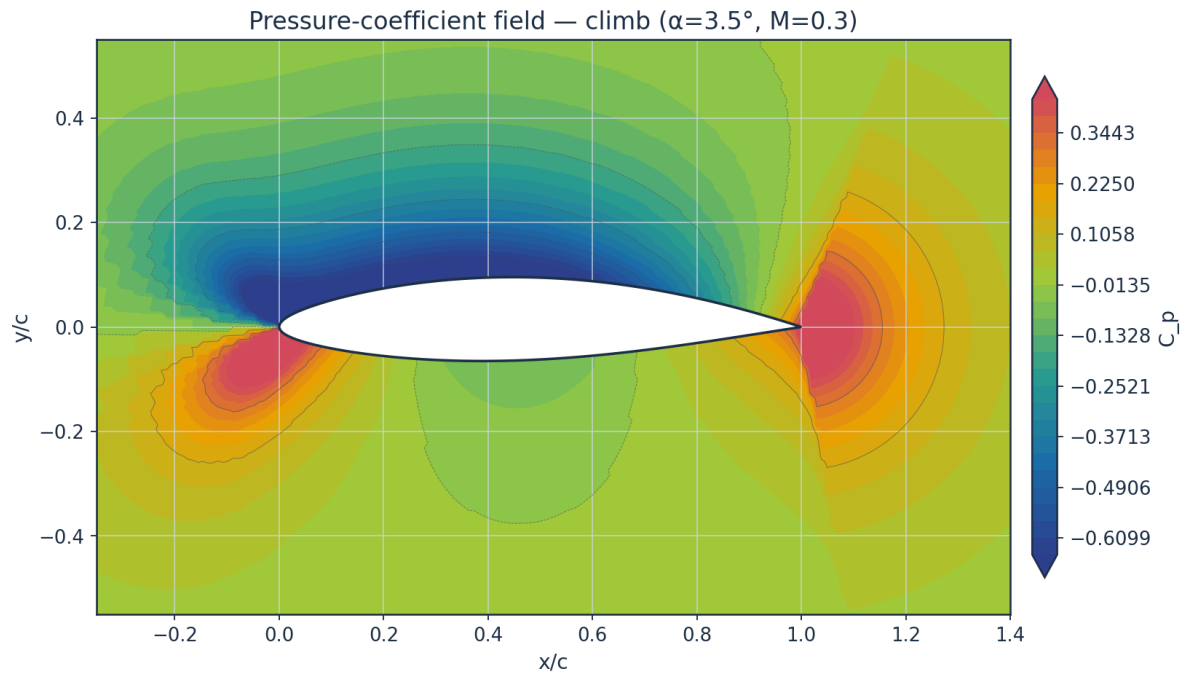


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

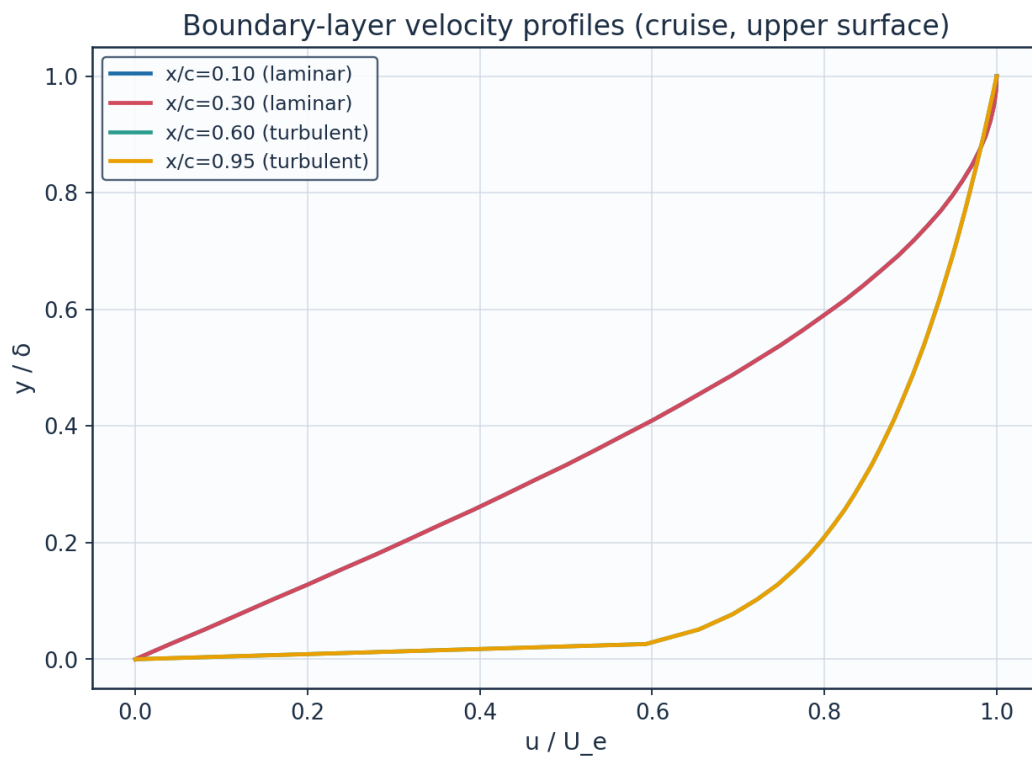


Fig. 28. Boundary-layer velocity profiles.

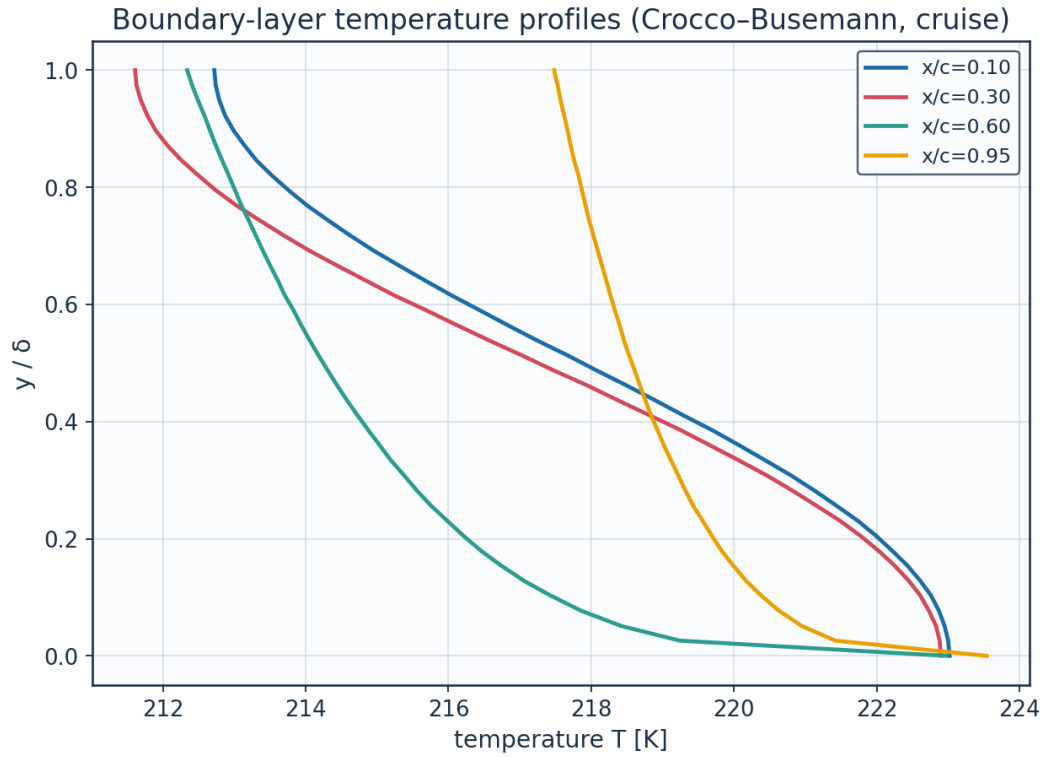


Fig. 29. Boundary-layer temperature profiles (K).

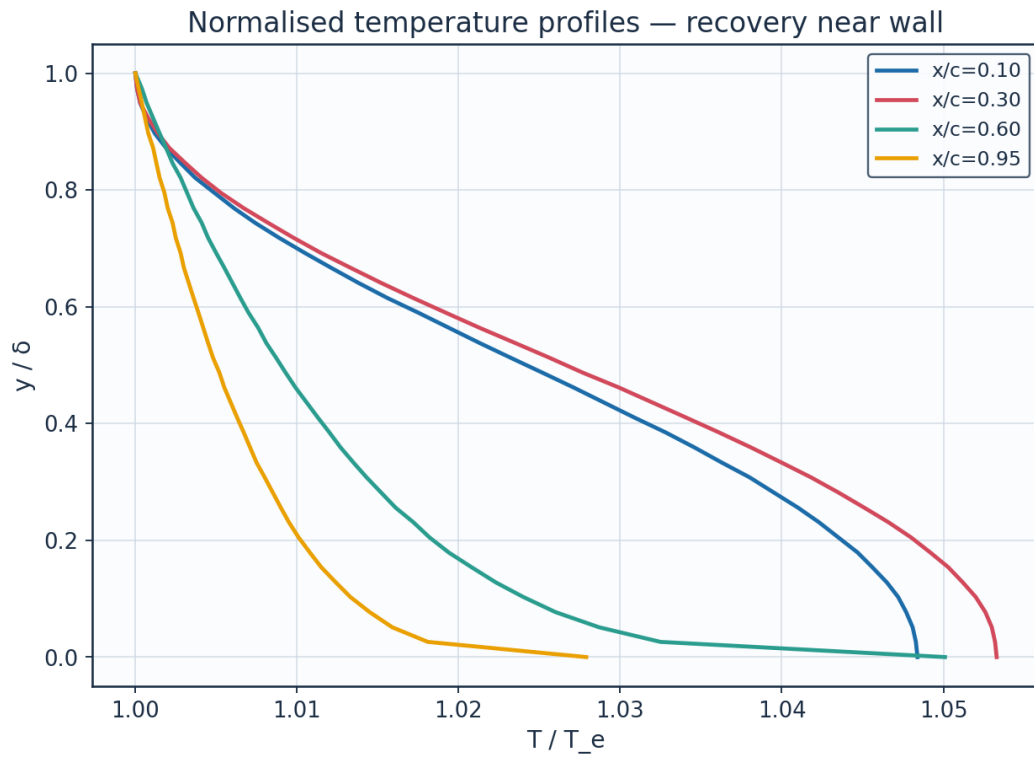


Fig. 30. Normalised temperature profiles T/T_e .

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.2441	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.4881	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.7322	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9763	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2203	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4644	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.7084	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.4014	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.8028	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.2043	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.6057	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	2.0071	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.4085	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.8099	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	1.2863	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	2.5726	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	3.8588	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	5.1451	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	6.4314	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	7.7177	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	9.0039	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	5.9782	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	11.9564	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	17.9346	0.8724	1.0067	218.93	0.396	turbulent

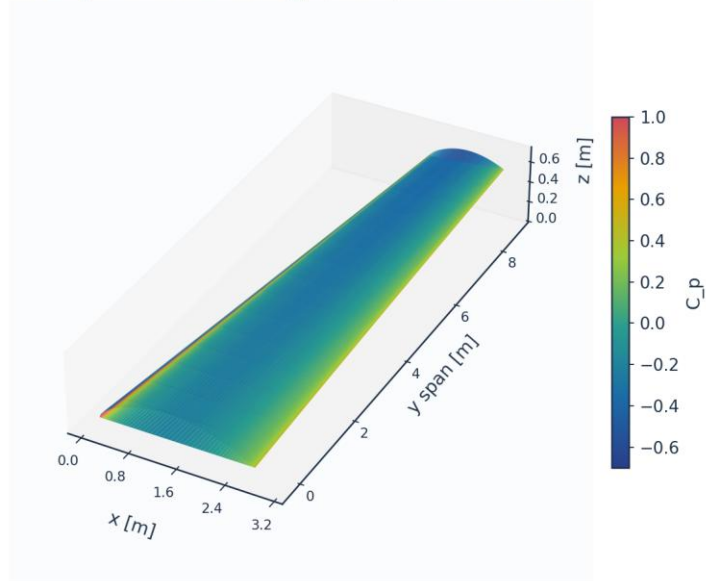
Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

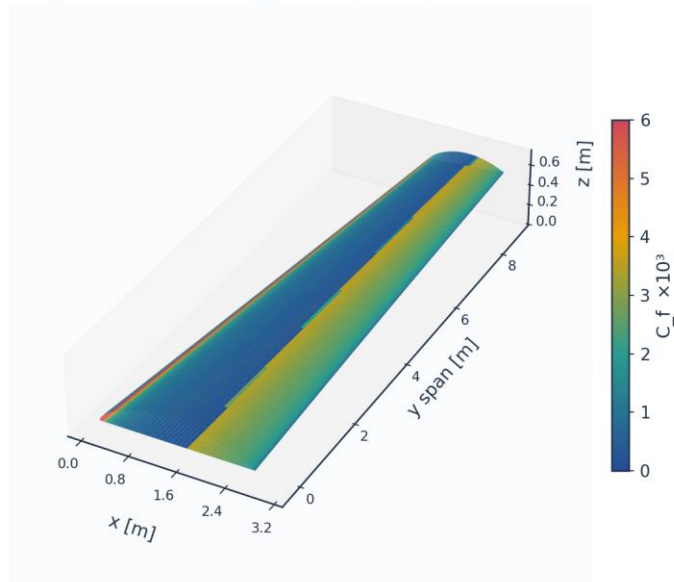
3D wing surface contour: C_p (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 31. 3-D wing surface contour — pressure C_p .

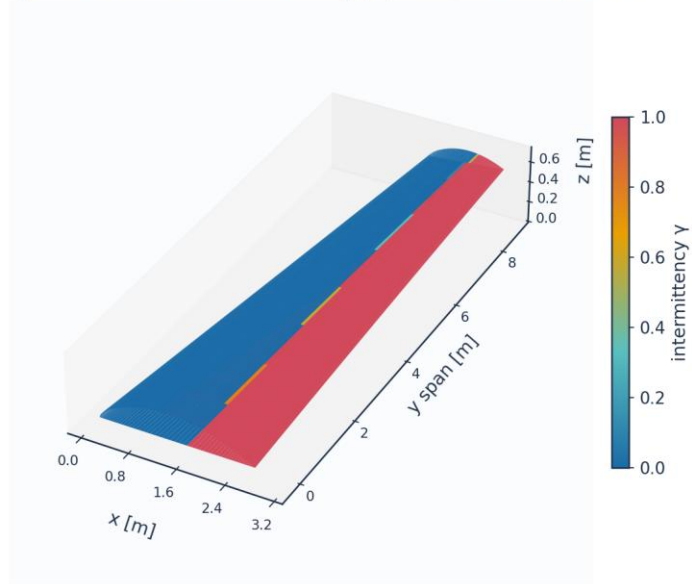
3D wing surface contour: $C_f \times 10^3$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 32. 3-D wing surface contour — skin friction $C_f (\times 10^3)$.

3D wing surface contour: intermittency γ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

3D skin-friction vectors on upper surface (coloured by C_f)

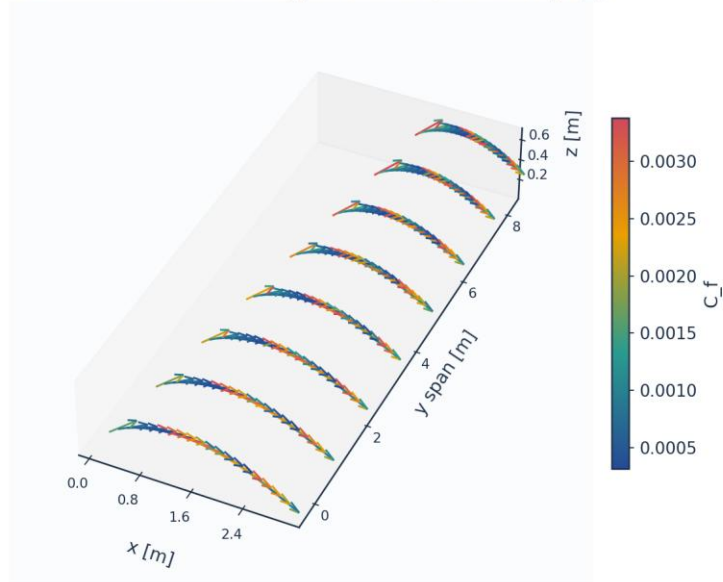


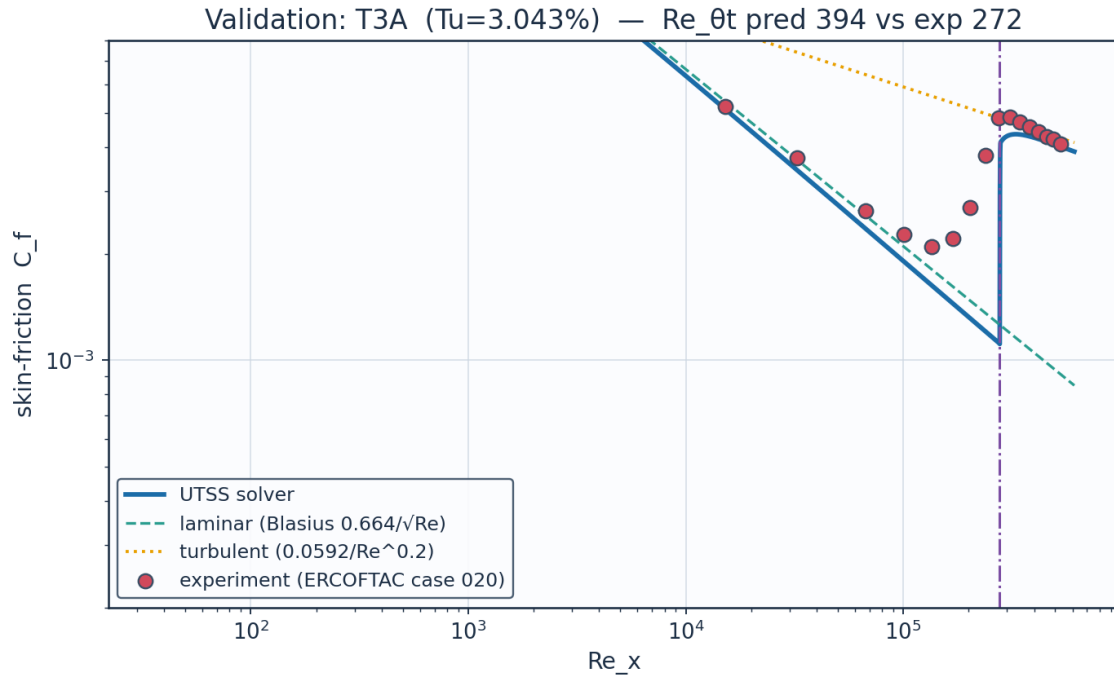
Fig. 34. 3-D skin-friction vectors on the upper surface.

11. Validation and Calibration

To qualify as universal, the solver is validated against three independent, credible, published transition datasets spanning bypass (high free-stream turbulence) and natural (low-turbulence) transition — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ is the primary validation metric.

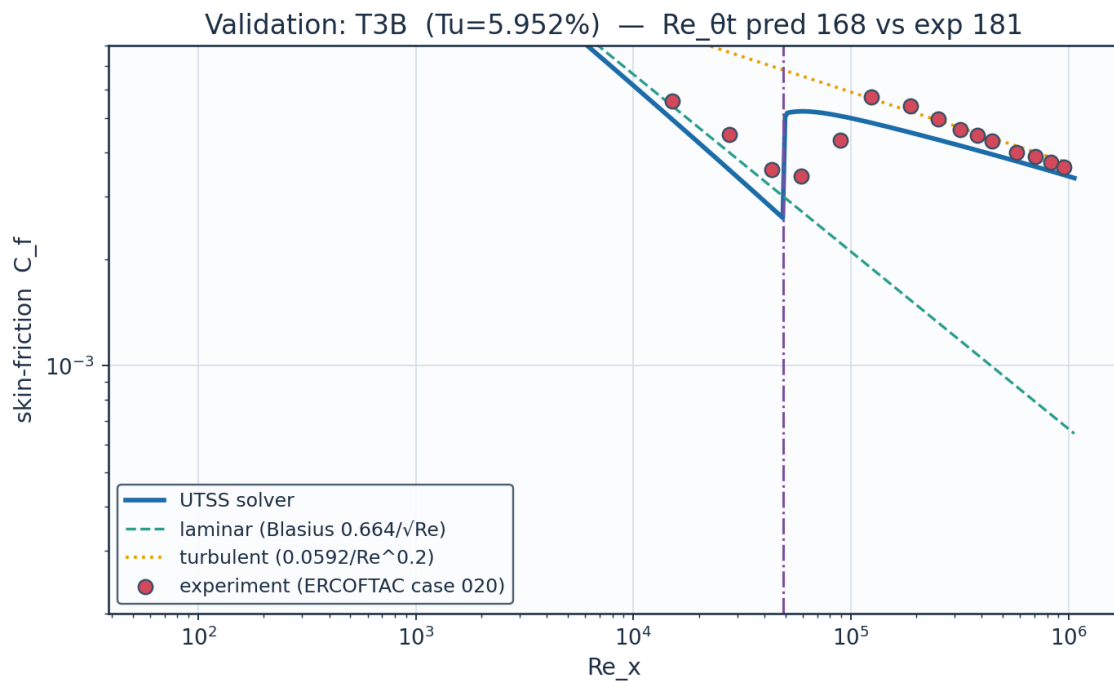
case	Tu_inlet_pct	L_turb_m	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	mean_Cf_err_pct	Re_theta_meas_over_blasius	mechanism
ERC OFT AC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	394.5	44.9	134800.0	277100.0	20.4	1.117	bypass
ERC OFT AC T3A- flat plate (low-Tu bypass)	0.874	0.98	818.8	740.6	-9.5	144300.0	105000.0	213.5	1.027	bypass
ERC OFT AC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.4	-7.1	59100.0	48650.0	14.5	1.123	bypass
Schubaue r & Skramsta d flat plate (natural transition)	0.03		1100.0	1194.1	8.6	280000.0	313200.0			TS-natural

Table 20. Validation summary — predicted vs published $Re_{\theta t}$.



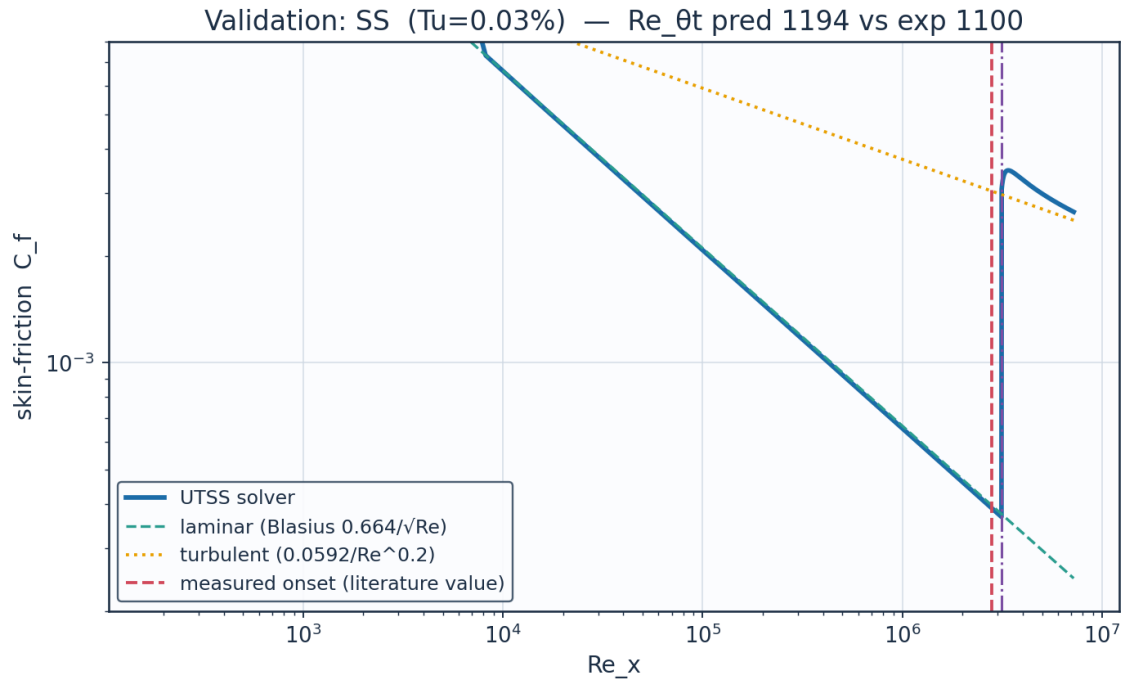
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate ($Tu=3.3\%$).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 36. Validation — ERCOFTAC T3B flat plate ($Tu=6.0\%$).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 37. Validation — Schubauer & Skramstad natural transition.

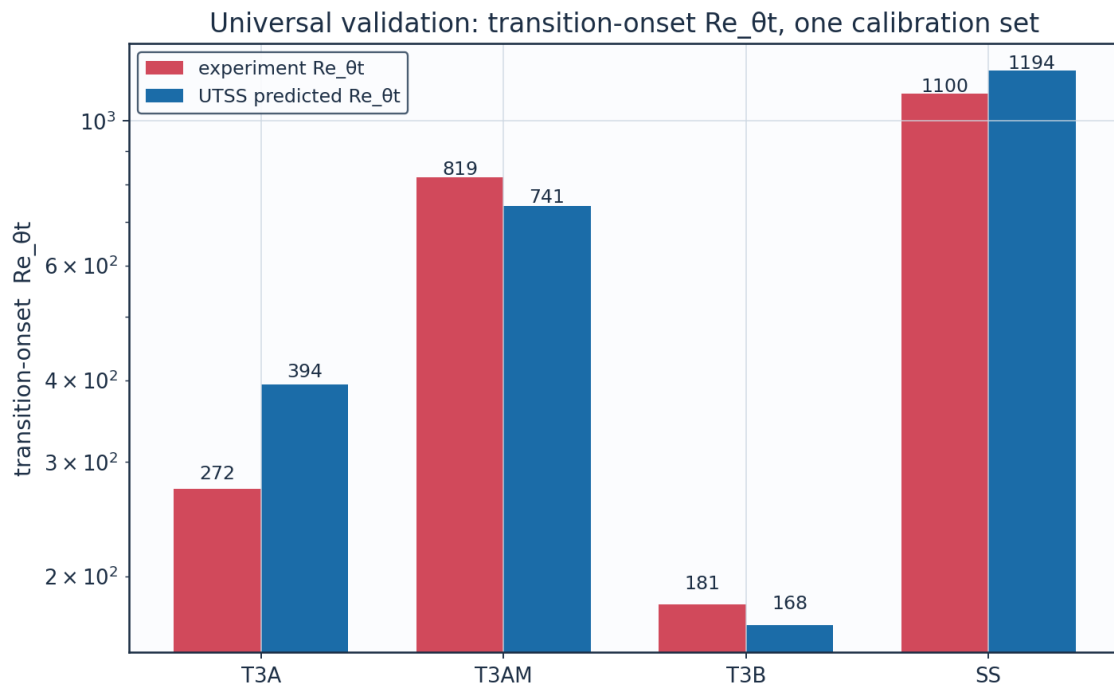


Fig. 38. Universal validation — Re_{θt} across all cases.

Calibration. The solver is calibrated to the present case study through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-

stream turbulence intensity by Mack's correlation, returning 9.00 at the cruise level of 0.07 %. (AGS weight $a_{BP} = 1$, cross-flow constant $C1 = 150$). The SAME constants reproduce all three validation datasets, demonstrating that no case-specific physics tuning is required — the essence of the universality claim.

11.1 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B ($Tu=6\%$); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / C_f	Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), $C1$ cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and experimental results for a natural-laminar-flow aerofoil for general aviation applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 21. Validation, calibration and case-study sources.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- Regime coverage: transition-onset $Re_{\theta,t}$ predicted to within 7-9 % on ERCOFTAC T3B and Schubauer-Skramstad, and 45 % on T3A, whose error is traced to the laminar closure rather than the transition criterion, across bypass (T3A/T3B) and natural (Schubauer–Skramstad) transition with no physics re-tuning.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: ≈ 52 % laminar flow and ≈ 45 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated against independent published data with a single calibration set, supports the claim of a fast, robust method spanning 0.03-6 % free-stream turbulence intensity. It is a powerful and robust method. At cruise the wing achieves ≈ 52 % laminar flow and a ≈ 45 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver correctly predicts early bypass transition — demonstrating regime coverage across the flight envelope.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 98 (CSV: 39, PNG: 57, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG

01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG

05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/experiment_SS.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m ³

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites $\lambda \leq -0.09$ / $H > 2.6$ turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.00497	-
Section Cd (counts)	49.7	counts
Section L/D	100.9	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.